

Brent Billingsley
Community Development Associate Director

Todd Williams
Community Development Deputy Director

Celeste Garza
Public Works Deputy Director

Christopher Wanamaker
County Engineer



PINAL COUNTY
WIDE OPEN OPPORTUNITY

Leo Lew
County Manager

Joe Ortiz
Development Services
Managing Director

**APPLICATION FOR CHANGE OF ZONING REGULATIONS IN AN UNINCORPORATED AREA OF
PINAL COUNTY, ARIZONA**
(All Applications Must Be Typed or Written in Ink)

Formal Zoning Change & Property Information:

(feel free to include answers and to these questions in a Supplementary Narrative, when doing so write see narrative on the space provided)

1. Pinal County Staff Coordinator: Sangeeta Deokar
 2. Date of Pre-application Review: 1 1 / 2 5 / 2 5 Pre-Application Review No.: Z-PA- 128-25
 3. Current Zoning (Please provide Acreage Breakdown): Suburban Ranch
 4. Requested Zoning (Please provide Acreage Breakdown): C2
 5. Parcel Number(s): 104-53-004H
 6. Parcel Size(s): 3.33 acres
 7. The existing use of the property is as follows: Vacant - undeveloped
 8. The exact use proposed under this request: Commercial Retail Center
 9. What is the Comprehensive Plan Designation for the subject property: _____
 10. Is the property located within three (3) miles of an incorporated community? ☐ YES ☒ NO
 11. Is an annexation into a municipality currently in progress? ☐ YES ☒ NO
 12. Is there a zoning violation on the property for which the owner has been cited? ☐ YES ☒ NO
- If yes, zoning violation # _____

13. Discuss any recent changes in the area that would support your application i.e.: zone change(s), subdivision approval, Planned Area Development (PAD), utility or street improvements, adopted comprehensive/area plan(s) or similar changes. See narrative

14. Explain why the proposed development is needed and necessary at this time. See narrative

INV#: _____ AMT: _____ DATE: _____ CASE: _____ Xref: _____

SUPPORTING INFORMATION

1. Note any services that are not available to the site. Discuss any improvements of services that would be paid for by the public: See narrative

2. What is the amount of traffic to be generated (# of trips/day, deliveries/week)? Show ingress/egress on the site plan: See narrative

3. How many parking spaces are to be provided (employees and customers)? Indicate these parking spaces on the site plan: See narrative

4. Is there a potential for excessive noise (I.E.; children, machinery) or the production of smoke, fumes, dust or glare with this proposed land use? If yes, how will you alleviate these problems for your neighbors?

See narrative

5. What type of landscaping are you proposing to screen this use from your neighbors?

See narrative

6. What type of signage are you proposing for the activity? Where will the signs be located?

See narrative

7. If the proposed land use involves any type of manufacturing or production process, provide a short synopsis of the processes utilizing diagrams, flowcharts and/or a short narrative: See narrative

8. Explain how the appearance and operation of the proposed land use will maintain the integrity and character of the zone in which the use is requested: See narrative

9. Have you discussed possible conditions that may be placed on the approval with the Planning Department?
☒ YES ☐ NO
10. Do you understand that if a condition is violated, that there is a public process by which your zoning may be reverted? ☒ YES ☐ NO

PROPERTY OWNERSHIP LIST

(Required for filing all applications)

See attached List

Instructions: Print Name, Address, City, State, Zip Code and Tax Parcel Number for each property owner within 600 feet of the subject parcel boundary. Feel free to attach a separate list if generated digitally. Please see "How to use the Buffer Tool" on our FAQ's page if you are generating the list.

Parcel No.: _____
 Name: _____
 Address: _____
 City/ST/Zip: _____

Parcel No.: _____
 Name: _____
 Address: _____
 City/ST/Zip: _____

Parcel No.: _____
 Name: _____
 Address: _____
 City/ST/Zip: _____

Parcel No.: _____
 Name: _____
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Parcel No.: _____
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Parcel No.: _____
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Parcel No.: _____
 Name: _____
 Address: _____
 City/ST/Zip: _____

Parcel No.: _____
 Name: _____
 Address: _____
 City/ST/Zip: _____

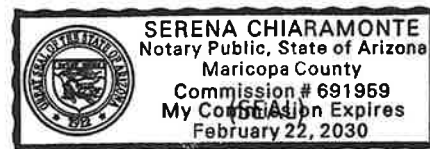
I hereby verify that the name list ^{attached} above was obtained on the 4th day of May, 2026, at the office of Pinal County Zoning GIS website and is accurate and complete to the best of my knowledge.
 (Source of Information)

On this 4th day of May, 2026, before me personally appeared Adam Baugh
 (Name of signor)

Signature A. Baugh Date 5/4/26

State of Arizona

County of Maricopa)ss.



My Commission Expires February 22nd, 2030

Signature of Notary Public Serena Chiaramonte

I certify the information included in this application is accurate, to the best of my knowledge. I have read the application and I have included the information, as requested. I understand if the information submitted is incomplete, this application cannot be processed. All notices will be sent to the applicant unless otherwise directed in writing

Withey Morris Baugh PLC	2121 E Highland Avenue Phoenix, Arizona 85016	
Name of Applicant	Address	
	hayes@wmbattorneys.com	602-230-0600
Signature of Applicant	E-Mail Address	Phone Number

Adam Baugh / Alex Hayes	2121 E Highland Avenue Phoenix, Arizona 85016	
Name of Agent/Representative	Address	
	adam@wmbattorneys.com	602 230 0600
Signature of Agent/Representative	E-Mail Address	Phone Number

The Agent/Representative has the authority to act on behalf of the landowner/applicant, which includes agreeing to stipulations. The agent will be the contact person for Planning staff and must be present at all hearings. Please use attached Agency Authorization form, if applicable.

If landowner is not the applicant, then applicant must submit a signed notarized consent form from the landowner with this application. Please use attached Consent to Permit form, if applicable.

AGENCY AUTHORIZATION

(To be completed by all landowners who do not represent themselves. Instructions for completing required information are in bold and brackets below lines. If applicant is a company, corporation, partnership, joint venture, trustee, etc., please use the corporate signature block and have the notary fill in the notarization section for corporations not individuals and cannot be submitted digitally)

TO: Pinal County Community Development
P.O. Box 2973
Florence, AZ 85232

TB Lone Star LLC & DB Lone Star LLC

[Insert Name -- If a Corporation, Partnership or Association, Include State of Incorporation]

Hereinafter referred to as "Owner," is/are the owner(s) of 3.33 acres located at
Gantzel Road and West Lone Star Road and further identified

[Insert Address of Property]

As assessor parcel number 104-53-004H and legally described as follows:

[Insert Parcel Number]

Insert Legal Description Here OR Attach as Exhibit A

Said property is hereinafter referred to as the "Property."

Owner hereby appoints Withey Morris Baugh, PLC

[Insert Agent's Name. If the Agent is a Company, Insert Company Name Only]

Hereinafter referred to as "Agent," to act on Owner's behalf in relation to the Property in obtaining approval from Pinal County for a minor land division and to file applications and make the necessary submittals for such approvals.

[Individual PROPERTY OWNER signature block and acknowledgment.

DO NOT SIGN HERE IF SIGNING AS AN OFFICER OF A CORPORATION SIGN NEXT PAGE]

[Signature]

2323 WEST UNIVERSITY DR, TUCSON, AZ 85728

[Address]

[Signature]

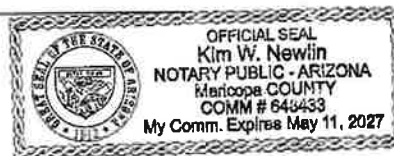
[Address]

Dated: 5/4/26

Dated: _____

STATE OF ARIZONA

COUNTY OF Maricopa) ss.



(SEAL)

The foregoing instrument was acknowledged before me, this 4th day May, 2026
by Tyson Breinholt

My Commission Expires May 11, 2027

Signature of Notary Public _____

Kim W Newlin
Printed Name of Notary

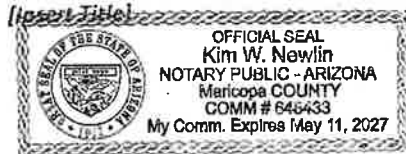
Kim W Newlin
Signature of Notary

CORPORATE PROPERTY OWNER SIGNATURE BLOCK AND ACKNOWLEDGMENT

TB Lone Star LLC & DB Lone Star LLC

By: Tyson Breakfast ^[Insert Company's or Trust's Name]
[Signature of Authorized Officer, or Trustee]
Its: Member Manager ^[Insert Title]

Dated: 5/4/26



STATE OF Arizona)
) ss.
COUNTY OF Maricopa)

The foregoing instrument was acknowledged before me, this may 4th day of May, 2026 by

Tyson Breakfast ^[Insert Signor's Name] ^[Insert Title]
TB Lone Star LLC & DB Lone Star LLC, Arizona, an,
^[Name of Company or Trust] ^[Insert State of Incorporation, if applicable]

And who being authorized to do so, executed the foregoing instrument on behalf of said entity for the purposes stated therein.

My Commission Expires: May 11, 2027

Kim W. Newlin
Printed Name of Notary

Kim W. Newlin
Signature of Notary

ALTERNATE: Use the following acknowledgment only when a second company is signing

On behalf of the owner:

STATE OF _____)
) ss. (Seal)
COUNTY OF _____)

The foregoing instrument was acknowledged before me, this _____ day of _____, 20____ by _____, who acknowledges himself/herself to be

^[Insert Signor's Name] _____, of _____
^[Title of Office Held] _____ ^[Second Company]

As _____ for _____, and who being
^[i.e. member, manager, etc.] ^[Owner's Name]

Authorized to do so, executed the foregoing instrument on behalf of said entities for the purposes stated therein.

My Commission Expires _____

Printed Name of Notary

Signature of Notary

Application Checklist:

A. Check the appropriate item:

☒ This Zone Change is being submitted without a PAD request

☐ This Zone Change is being submitted in conjunction with a PAD request.

Include the Development Agreement, if applicable. (Check mark on page 9, line item I of the application check list.)

The applicant must complete a PAD application. – *(Please utilize the "PAD Book" and the "Site Plan" of the PAD application to fulfill the Zoning Application "Narrative" and "Site Plan" in lieu of while having separate copies for each application).*

B. Hold a Neighborhood/Community Meeting:

☒ Notify all property owners within 1200' (feet)

☒ Hold the meeting within five (5) miles of the subject property

☒ Hold the meeting between 5:00 pm – 9:00 pm

☒ Include with the application the following:

- Copy of Notice of Neighborhood/Community Meeting
- List of property owners notified - *(Use page 5 of this application)*
- Minutes of the meeting
- Attendance sign-in sheet with names & addresses

C. Submit a completed "Agency Authorization" form *(if applicable)*.

D. Submit a written Narrative concerning the proposed development *(if not submitting in conjunction with a PAD Application)* to include:

☒ 1. Title Page

☒ 2. Purpose of Request

☒ 3. Description of Proposal

- a. Nature of the Project including Proposed Land Use
- b. Conformance to adopted Comprehensive Plan
- c. Answers to the questions from the **Supporting Information** sheet
- d. Location & Accessibility
- e. Utilities & Services
- f. Neighborhood Meeting Information
- g. Existing Zoning Stipulation
- h. Appendix, as applicable

E. **Submit a Site Plan** (*if not submitting in conjunction with a PAD Application*). The submittal shall be professionally prepared (*by a surveyor, architect, or other design professional*) and drawn at a sufficient scale as to not exceed a print size larger than 11" X 17". The lettering shall be of sufficient size to be legible when reduced to an 8½" X 11" print and include:

- ☒ 1. Legal Description of total site.
- ☒ 2. Name(s) of Landowner(s), Developer, Applicant and Person or Firm preparing plan.
- ☒ 3. North Arrow, Scales (*written and graphic*), Preparation Date and Subsequent Revision Dates.
- ☒ 4. Location of all Existing & Proposed Structures & Buildings
- ☒ 5. Location of all Existing & Proposed Utilities with Location & Width of Associated Easements.
- ☒ 6. All Existing & Proposed Public and/or Private Streets with Location & Width of Associated Easements & Right-of-Ways.
- ☒ 7. All Points of Ingress & Egress.
- ☒ 8. Location & Types of Existing & Proposed Landscaping.
- ☒ 9. Indicate Location, Type, Height, & Materials for Proposed Walls, Fences & Signs.
- ☒ 10. Show whether the property is adjacent to a projected regionally significant route (RSR) as identified on the Corridor Preservation Map (Figure 9) in the Regionally Significant Routes for Safety and Mobility (RSRSM), Final Report. If adjacent to a projected RSR, show how applicant will comply with the RSRSM Final Report and the RSRSM Access Management Manual.

F. **Submit the following information regarding Water Supply:**

- ☒ 1. Identify the water service provider
- ☒ 2. Provide an estimated range of water demand and include an explanation of the method used to obtain the estimate
- ☒ 3. Provide information about water supply and source, including renewable and non-groundwater supplies
- ☒ 4. If a portion of the water supply for the proposed project is groundwater, the application shall be accompanied by the following information that is publicly available from the Arizona Department of Water Resources (ADWR) and/or Arizona Geological Survey, or otherwise available to the land owner:
 - o Depth to bedrock & Depth to groundwater
 - o Known fissures or land subsidence in the area
 - o Known wells in the area, available information on status and water levels
 - o Summary of data-gathering efforts and sources of information

G. Provide a copy of the **Title Report**

H. **Submit** a copy of a certified A.L.T.A. survey, including a legal description of proposed zoning districts.

I. **Development Agreements (DA):**

☐ Provide a copy of the approved DA, if applicable.

J. **Submit a Drainage report and a TIA report.** *(Public Works fees for these reviews will be due at the time of submittal and each subsequent review until deemed acceptable).*

K. **Aware** that earth fissure maps are available online from the Arizona State Geologic Survey.

L. **Submit a list of all property owners within 600' (feet)** of the subject property boundary showing name, mailing address and tax parcel numbers. This list must be obtained within 30 days prior to application submission. A map showing the 600' boundary and parcels must be included as well (A Tax Assessor Parcel Map is Acceptable). - *(This list is a separate list from the "Neighborhood/Community Meeting list of 1,200' however use Page 5 of this application as well).*

M. **Complete and Submit** the "Comprehensive Plan Compliance Checklist" a copy of the Checklist can be found here:
<http://www.pinalcountyz.gov/CommunityDevelopment/Planning/Documents/Planning%20Applications/CompPlan%20Checklist.pdf>

N. **Submit** the Non-Refundable fees for a zone change outlined on page one of the Zone Change Application.

O. **Submit** all documentation outlined in the Zone Change application in a multi-PDF format per item of the application with all supporting documentation via the online portal site at:
<https://citizenaccess.pinalcountyz.gov/CitizenAccess/>. *

- a. An ESRI shapefile for land use (conceptual) which shows all proposed zoning lines and zoning classifications for the project in NAD_1983_stateplan_arizona_central_fips_0202_intlfeet projection

** Your application must be submitted digitally via the online portal site. Please call or email the Planning Division for more information.*

P. **Aware to Install Broadcast Notification Sign(s) on the site in conformance with the information shown in this application.** *(See page 10 & 11 of this application for illustrative details).* **Aware** that newspaper advertising fees and postage must be paid **by the applicant.** *(in addition to application fees*

Q. Signature at the end of the "Checklist" stating you have reviewed and addressed all areas within it.

I certify that I have submitted all the required information listed above, and I understand that this application for a Zone Change cannot be processed until all required information is submitted.

Adam Baugh

2121 E Highland Avenue Phoenix, Arizona 85016

Name of Applicant

Address



Signature

5/4/26

Date

TITLE REPORT SCHEDULE 'B' ITEMS:

1. TAXES AND ASSESSMENTS collectible by the County Treasurer, a lien not yet due and payable for the following year:
2025 – NOT SURVEY ITEM
2. LIABILITIES AND OBLIGATIONS imposed upon said land by reason of its inclusion within the following named District:
a. Central Arizona Water Conservation District
b. Pinal County Flood Control District – NOT SURVEY ITEM
3. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Document No: Book 84 of Deeds, Page 97
Purpose: Public Highway
4. THE FOLLOWING MATTERS disclosed by following instrument recorded in:
Entitled: Resolution Ordering Establishment of a County Road
Recorded: August 06, 1957
Document No: Docket 184, Page 299
5. MATTERS SHOWN ON SURVEY recorded in Book 1 of Surveys, Page 50. – AS SHOWN ON SURVEY
6. RESTRICTIONS, CONDITIONS, COVENANTS, RESERVATIONS, LIABILITIES AND OBLIGATIONS, including but not limited to any recitals creating easements or party walls, omitting, if any, from the above, any restrictions based on race, color, religion, sex, handicap, familial status or national origin contained in instrument:
Recorded in Document No: Docket 912, Page 983
Re-Recorded in Document No: Docket 913, Page 674
Amendment of said covenants, conditions and restrictions:
Fee No: 1997–033615
Complete Amendment of said covenants, conditions and restrictions:
Fee No: 2000–043364
First Amendment to Complete Amendment of said covenants, conditions and restrictions:
Fee No: 2020–1066222
Termination of Easement Segment of said covenants, conditions and restrictions:
Fee No: 2022–025224
Liens and charges as set forth in the above mentioned declaration,
Payable to: Sun Valley Farms IV Property Owners Association, an Arizona non–profit corporation – TERMINATED
7. RESTRICTIONS, CONDITIONS, COVENANTS, RESERVATIONS, including but not limited to any recitals creating easements, liabilities, obligations or party walls, omitting, if any, from the above, any restrictions based on race, color, religion sex, handicap, familial status or national origin contained in instrument:
Recorded in Fee No: 1996–021624 – AFFECTS, NOT MAPPABLE
8. THE EFFECT of Ordinance No. 122000–BS by the Board of Supervisors of Pinal County, Arizona recorded in:
Recorded: January 05, 2001
Fee No: 2001–000756
Concerning: Relating to Residential Fireplace Restrictions
Thereafter, Ordinance No. 121207–A01 recorded in Fee No. 2008–001862. – AFFECTS, NOT MAPPABLE
9. MATTERS SHOWN ON SURVEY recorded in Book 5 of Surveys, Page 281.
10. MATTERS SHOWN ON SURVEY recorded in Book 6 of Surveys, Page 17. – AS SHOWN ON SURVEY
11. AGREEMENT, according to the terms and conditions, contained therein:
Entitled: Road Maintenance Agreement
Recorded: January 28, 2002
Fee No: 2002–003998 – AFFECTS, NOT MAPPABLE
12. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Fee No: 2002–003999
Re–Recorded in Fee No: 2002–038417
Purpose: Salt River Project Electric – AS SHOWN ON SURVEY
13. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Fee No: 2002–004000
Re–Recorded in Fee No: 2002–038418
Purpose: Ingress, Egress and Public Utilities – AS SHOWN ON SURVEY
14. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Fee No: 2002–004001
Purpose: Irrigation – AS SHOWN ON SURVEY
15. MATTERS SHOWN ON SURVEY recorded in Book 7 of Surveys, Page 54.
16. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Fee No: 2003–000146
Purpose: Transmission and Distribution of Electricity – AS SHOWN ON SURVEY
17. RESTRICTIONS, CONDITIONS, COVENANTS, RESERVATIONS, including but not limited to any recitals creating easements, liabilities, obligations or party walls, omitting, if any, from the above, any restrictions based on race, color, religion sex, handicap, familial status or national origin contained in instrument:
Recorded in Fee No: 2003–024801 – AFFECTS, NOT MAPPABLE
18. THE FOLLOWING MATTERS disclosed by following instrument recorded in:
Entitled: Acknowledgement and Acceptance
Recorded: May 14, 2004
Fee No: 2004–035651 – AFFECTS, NOT MAPPABLE
19. RESTRICTIONS, CONDITIONS, COVENANTS, RESERVATIONS, LIABILITIES AND OBLIGATIONS, including but not limited to any recitals creating easements or party walls, omitting, if any, from the above, any restrictions based on race, color, religion, sex, handicap, familial status or national origin contained in instrument:
Recorded in Fee No: 2004–068355
Liens and charges as set forth in the above mentioned declaration,
Payable to: Sun Valley Farms IV Red Fern Irrigation Association – AFFECTS, NOT MAPPABLE
20. THE EFFECT of Resolution No: 111004–IGR by the Board of Supervisors of Pinal County, Arizona recorded in:
Recorded: November 16, 2004
Fee No: 2004–093152
Concerning: Establish a County Highway Ironwood/Gantzel Road – AFFECTS, NOT MAPPABLE
21. EASEMENT and rights incident thereto, as set forth in instrument:
Recorded in Fee No: 2005–067011
Purpose: Roadway – AS SHOWN ON SURVEY
22. AGREEMENT, according to the terms and conditions, contained therein:
Entitled: Agreement to Donate Easement and Waiver of Appraisal and Compensation
Recorded: June 06, 2005
Fee No: 2005–067012 – AFFECTS, NOT MAPPABLE
23. MATTERS SHOWN ON SURVEY recorded in Fee No. 2024–044725. – AS SHOWN ON SURVEY

IF EXTENDED COVERAGE IS REQUESTED, THE FOLLOWING BELOW EXCEPTIONS WILL BE ADDED:

24. Reservations contained in the Patent.
From: State of Arizona
Recording: February 28, 1946
Document No: Book 76 of Deeds, Page 61
25. WATER RIGHTS, claims or title to water, and agreements, covenants, conditions or rights incident thereto, whether or not shown by the public records.
This exception is not limited by reason of the disclosure of any matter relating to Water Rights as may be set forth elsewhere in Schedule B. – NOT SURVEY ITEM
26. RIGHTS OF PARTIES in possession.
NOTE: This exception will be amended or deleted upon the submission of the corresponding documents required in Schedule B, Part I.
27. INTENTIONALLY OMITTED.
28. LOCATION OF IMPROVEMENTS, easements, discrepancies, conflicts in boundary lines, shortage in area, encroachments or any other facts which a correct survey would disclose.
NOTE: This exception will be amended or deleted upon the submission of the corresponding documents required in Schedule B, Part I.

ALTA/NSPS LAND TITLE SURVEY

A PORTION OF SECTION 30, TOWNSHIP 2 SOUTH, RANGE 8 EAST OF THE GILA AND SALT RIVER BASE AND MERIDIAN, PINAL COUNTY, ARIZONA

LEGAL DESCRIPTION :

THE LAND REFERRED TO HEREIN BELOW IS SITUATED SAN TAN VALLEY, IN THE COUNTY OF PINAL, STATE OF ARIZONA, AND IS DESCRIBED AS FOLLOWS:

BEGINNING at the Northeast corner of Parcel 4, of SUN VALLEY FARMS UNIT IV, according to Book 1 of Surveys, Page 50, records of Pinal County, Arizona, being a portion of Section 30, Township 2 South, Range 8 East of the Gila and Salt River Base and Meridian, Pinal County, Arizona, having a basis of bearing of South 00 degrees 17 minutes 22 seconds West along the Easterly line of the Northeast quarter of said Section 30;

Thence South 00 degrees 17 minutes 22 seconds West along said Easterly line of the Northeast quarter of Section 30, a distance of 584.10 feet to a point from which the Southeast corner of said Parcel 4 bears South 00 degrees 17 minutes 22 seconds West, 674.81 feet distant therefrom;

Thence North 89 degrees 38 minutes 38 seconds West, parallel with the South line of said Parcel 4, a distance of 246.02 feet;

Thence North 00 degrees 17 minutes 22 seconds East, parallel to the East line of said Parcel 4, a distance of 584.49 feet to a point on the North line of said Parcel 4, from which point the Northwest corner of said Parcel 4 bears North 89 degrees 33 minutes 07 seconds West, measured (North 89 degrees 32 minutes 09 seconds West, record), 1060.99 feet therefrom;

Thence South 89 degrees 33 minutes 07 seconds East, measured (South 89 degrees 32 minutes 09 seconds East, record), along the North line of said Parcel 4, a distance of 246.02 feet to the POINT OF BEGINNING.

BASIS OF BEARING:

THE EASTERLY LINE OF THE NORTHEAST QUARTER OF SECTION 30, TOWNSHIP 2 SOUTH, RANGE 8 EAST, GILA AND SALT RIVER MERIDIAN. PINAL COUNTY, ARIZONA BEARS SOUTH 00°17'22" WEST.

SITE INFORMATION:

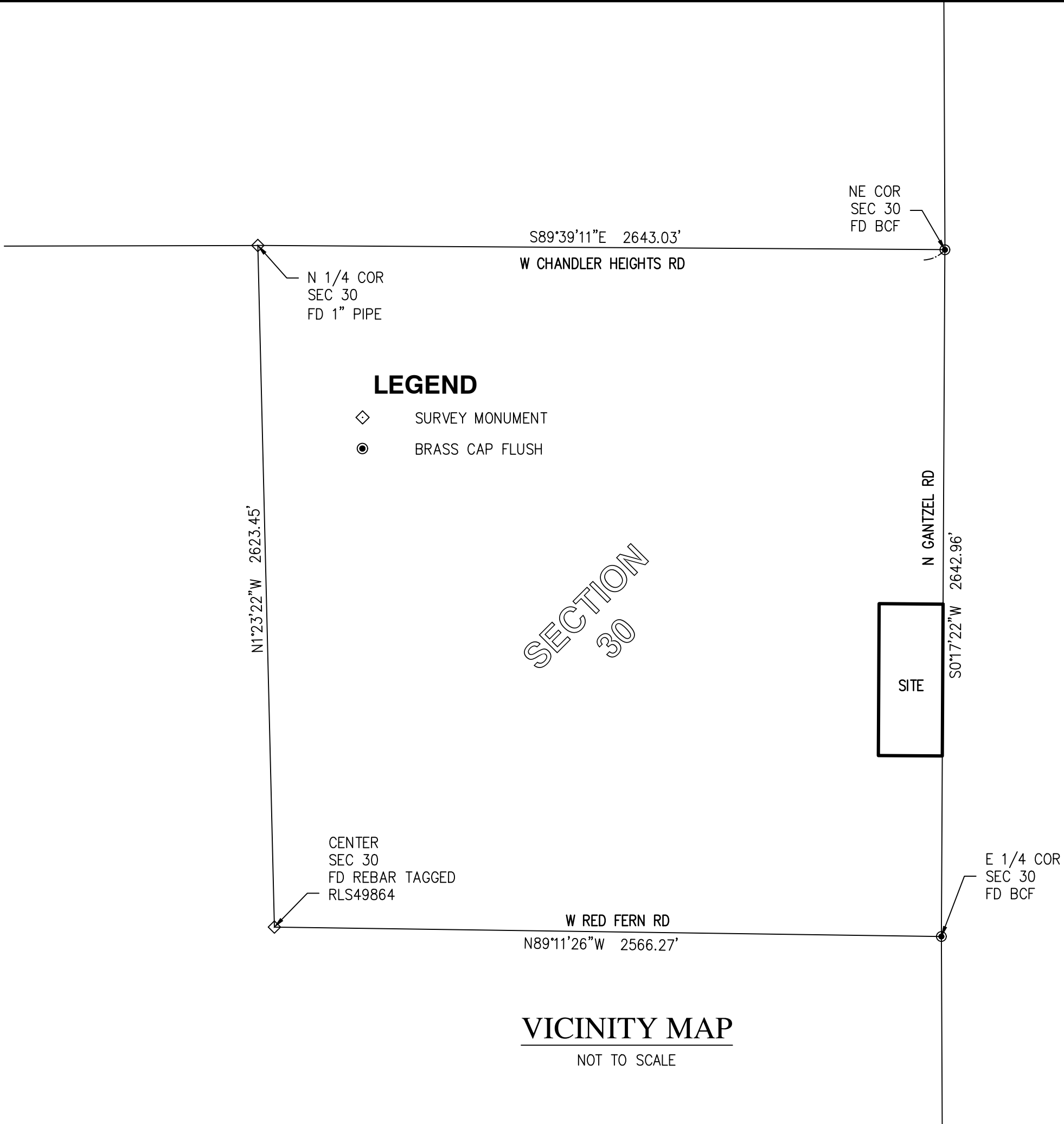
PARCEL	APN:	AREA
PARCEL	104–53–004H	143,748 SF OR 3.300 AC GROSS
TOTAL		143,748 SF OR 3.300 AC GROSS

OWNER :
ARUMALLA PROPERTIES LLC

ZONING INFORMATION:

PARCEL	ZONING:	REQUIREMENT:
PARCEL	SUBURBAN RANCH	MAXIMUM BUILDING HEIGHT – 30' MINIMUM LOT AREA – 144,000 SF (3.30 AC) MINIMUM AREA PER DWELLING UNIT – 144,000 SF (3.30 AC) MINIMUM FRONT YARD – 50 FEET MINIMUM SIDE YARD – 10 FEET MINIMUM REAR YARD – 50 FEET

BY THE COUNTY OF PINAL ZONING ORDINANCE NO. 61862, DATED 09/19/2025.



SHEET INDEX:

SHEET 1	COVER, NOTES, LOCATION MAP
SHEET 2	SURVEY, LEGEND

TITLE REPORT:

THE BOUNDARY SURVEY OF THE SUBJECT PROPERTY AS SHOWN HEREON WAS DETERMINED FROM EXHIBIT "A" IN THE CONDITION OF TITLE REPORT ISSUED BY LANDMARK TITLE ASSURANCE AGENCY OF ARIZONA LLC, FILE No. 06208618–026–NM2, DATED FEBRUARY 24, 2025, AND IS LISTED HERE UNDER THE HEADING "LEGAL DESCRIPTION"

NOTES:

1. EXCEPT AS NOTED, SURVEYOR HAS MADE NO INDEPENDENT INVESTIGATION OR SEARCH FOR EASEMENTS OF RECORD, ENCUMBRANCES, RESTRICTIVE COVENANTS, OWNERSHIP TITLE EVIDENCE, OR ANY OTHER MATTER THAT THE PROVIDED TITLE REPORT MAY DISCLOSE.
2. DECLARATION IS MADE TO ORIGINAL PURCHASER ONLY. SURVEY IS NOT TRANSFERABLE TO SUBSEQUENT OWNERS OR ADDITIONAL INSTITUTIONS WITHOUT WRITTEN PERMISSION OF THE SURVEYOR.
3. SURVEY IS VALID ONLY IF PRINT BEARS SEAL AND/OR SIGNATURE OF SURVEYOR.
4. FIELDWORK PERFORMED DURING SEPTEMBER 2025.
5. CONDITIONS SHOWN ARE THOSE THAT EXISTED AT THE TIME OF THE SURVEY.
6. ALL BOUNDARY DIMENSIONS SHOWN ARE MEASURED (M) UNLESS OTHERWISE NOTED.
7. UTILITIES SHOWN ARE BASED ON BLUE STAKE, FIELD COLLECTED DATA, CONSTRUCTION DOCUMENTS AND TOWN OF SAN TAN VALLEY AS–BUILT PLANS. NO UNDERGROUND INVESTIGATION WAS PERFORMED.
8. NO PROPOSED STREET RIGHT–OF–WAY CHANGES WERE MADE AVAILABLE TO THIS SURVEYOR. NO EVIDENCE OF RECENT STREET OR SIDEWALK CONSTRUCTION WERE OBSERVED DURING THE SURVEY.
9. THE WORD CERTIFY AS SHOWN OR USED HEREON MEANS AN EXPRESSION OF PROFESSIONAL OPINION REGARDING THE FACTS OF THE SURVEY AND DOES NOT CONSTITUTE A WARRANTY OR GUARANTEE EXPRESSED OR IMPLIED.
10. NO EVIDENCE OF RECENT EARTH MOVING, BUILDING CONSTRUCTION OR ADDITIONS WAS OBSERVED IN THE PROCESS OF THE SURVEY.

SURVEYOR'S CERTIFICATION:

CERTIFIED TO:
PUKJ INVESTMENTS, LLC, AN ARIZONA LIMITED LIABILITY COMPANY
FIRST FIDELITY BANK
ARUMALLA PROPERTIES, LLC, AN ARIZONA LIMITED LIABILITY COMPANY

THIS IS TO CERTIFY THAT THIS MAP OR PLAT AND THE SURVEY ON WHICH IT IS BASED WERE MADE IN ACCORDANCE WITH THE 2021 MINIMUM STANDARD DETAIL REQUIREMENTS FOR ALTA/NSPS LAND TITLE SURVEYS, JOINTLY ESTABLISHED AND ADOPTED BY ALTA AND NSPS, AND INCLUDES ITEMS 1, 8, 11, 16, 17, AND 18 OF TABLE A THEREOF. THE FIELDWORK WAS COMPLETED ON SEPTEMBER 17, 2025.

DATE OF PLAT OR MAP: OCTOBER 21, 2025.

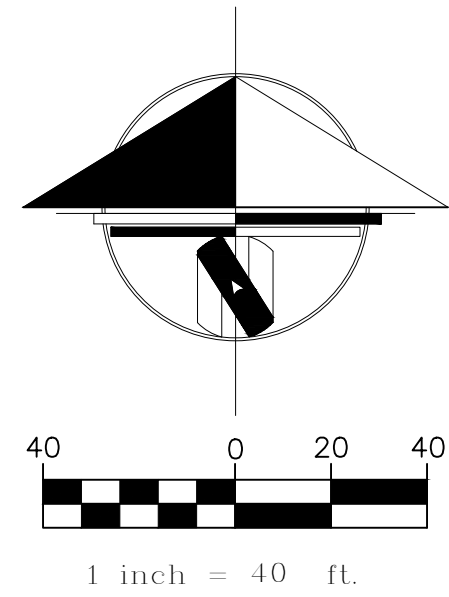
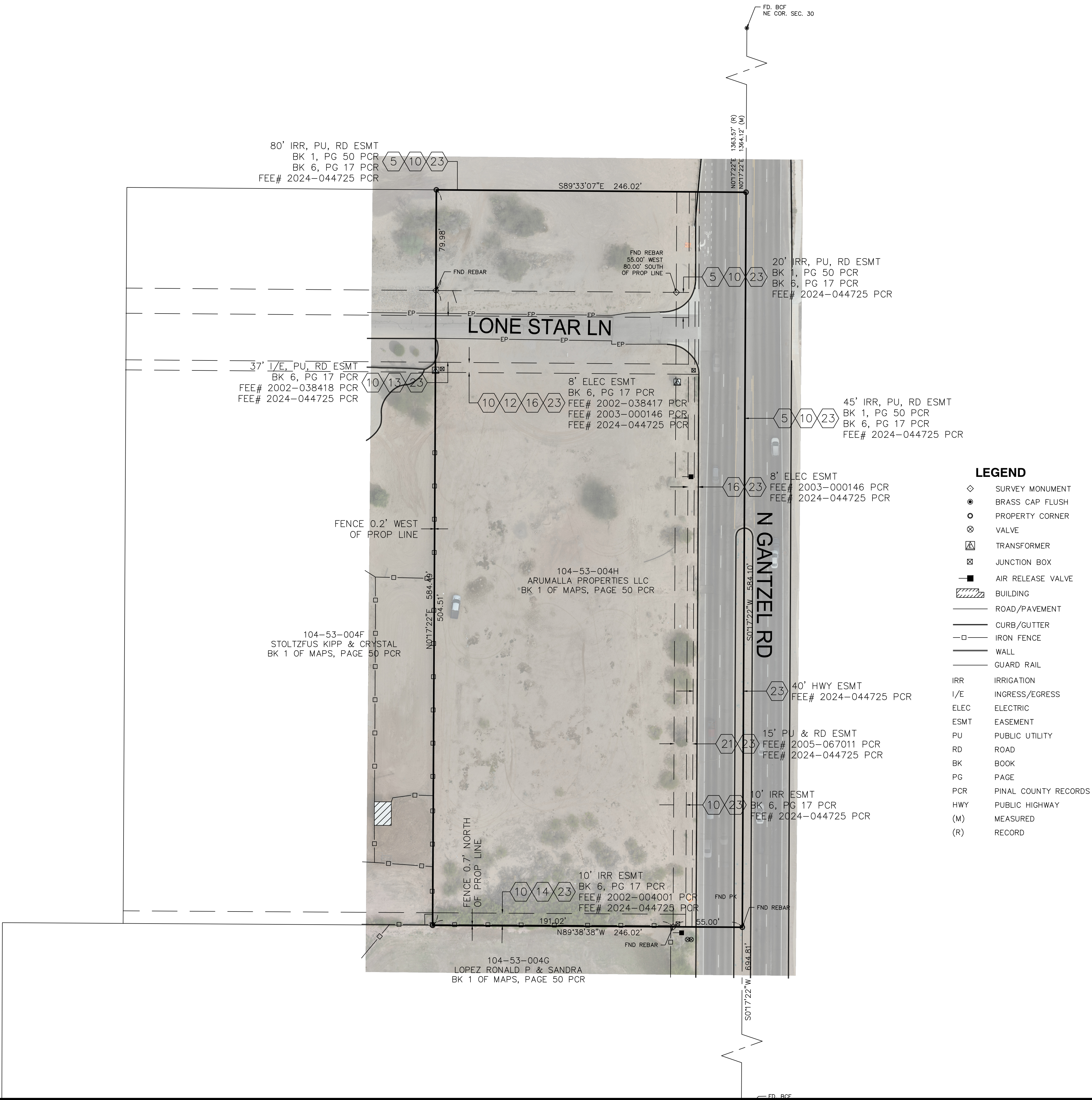
ROBERT S. UNGER
ARIZONA R.L.S. 35306



SYNERGY GEOMATICS
2125 E. 5th STREET, #101
TEMPE, AZ 85281
PH: (480) 406-3648
EMAIL: rob@syn-geo.com



DRAWN: TG	CHECKED: RU	DATE: 10/21/25	DWC: 25077A01.dwg
ALTA/NSPS SURVEY	15 W LONE STAR LN	SAN TAN VALLEY, AZ	PROJECT NO. 25_077 SHEET NUMBER 1 OF 2



LEGEND

- ◇ SURVEY MONUMENT
- BRASS CAP FLUSH
- PROPERTY CORNER
- ⊗ VALVE
- ⊠ TRANSFORMER
- ⊞ JUNCTION BOX
- AIR RELEASE VALVE
- ▨ BUILDING
- ROAD/PAVEMENT
- CURB/GUTTER
- IRON FENCE
- WALL
- GUARD RAIL
- IRR IRRIGATION
- I/E INGRESS/EGRESS
- ELEC ELECTRIC
- ESMT EASEMENT
- PU PUBLIC UTILITY
- RD ROAD
- BK BOOK
- PG PAGE
- PCR PINAL COUNTY RECORDS
- HWY PUBLIC HIGHWAY
- (M) MEASURED
- (R) RECORD



PROJECT NO. 25_077	SHEET NUMBER 2 OF 2	SYNERGY GEOMATICS 2125 E. 5th STREET, #101 TEMPE, AZ 85281 PH: (480) 406-3648 EMAIL: rob@syn-geo.com		REV.	DATE	BY
		ALTA/NSPS SURVEY 15 W LONE STAR LN SAN TAN VALLEY, AZ				



NEIGHBORHOOD PUBLIC NOTICE

Pinal County Rezoning Request

Parcel 104-53-004H

Southwest Corner of Gantzel Road & Lone Star Lane

Dear Property Owner or Resident:

This notice is being provided in accordance with Pinal County public participation requirements for rezoning applications. A request has been submitted to rezone the vacant property located at the southwest corner of Gantzel Road and Lone Star Lane, identified as Assessor's Parcel Number 104-53-004H. The subject property is approximately 3.3 acres in size and is currently zoned Residential Suburban Ranch (RSR).

The applicant is requesting to rezone the property to Commercial C-2 with a PAD overlay that will restrict some uses. The proposed zoning would allow development of a neighborhood-scale commercial center consisting of two 10,000-square-foot buildings. Anticipated uses include retail establishments, professional office tenants, restaurants, and neighborhood-serving personal services such as barbershops.

Properties located west and south of the subject parcel are currently zoned Residential Suburban Ranch. As part of the required public outreach process, nearby residents and property owners are invited to attend a neighborhood meeting to review the proposal, ask questions, and provide comments directly to the property owner and project representatives.

See other side of this notification for subject property location map and neighborhood meeting location/time information.



Subject Property Location Map

Neighborhood Meeting Information

Location:

**QC Library, Zane Grey Room
21802 S. Ellsworth Road,
Queen Creek, AZ 85142**

Date: February 26, 2026

Time: 5:00 p.m. – 9:00 p.m.

Representatives for the applicant will be available throughout the meeting to present the proposed rezoning, discuss anticipated site improvements, and address questions regarding potential impacts, compatibility, and the overall development intent.

Your participation is encouraged. This notice is being provided to ensure that surrounding property owners are informed of the request prior to formal consideration by Pinal County.

Sincerely,

Greg Bandelin – 480-947-6844
Sixty First Place Architects
3200 N. Hayden Rd., Suite 285
Scottsdale, AZ 85251

on
com
oo.com
t.com
yahoo.com
@yahoo.com
com

Neighborhood Rezoning Information Sheet

Proposed Commercial Use Center – Gantzel Road

Project Overview

The property owner is requesting to rezone a 3.3-acre vacant parcel along Gantzel Road from Suburban Ranch (SR) to Commercial Use Center. The goal is to create a small-scale commercial center that provides everyday services for nearby residents.

What the Rezoning Would Allow

- Retail shops
- Personal and professional services (hair, nails, small offices)
- Indoor-dining restaurants (outdoor patios and a future drive-through possible)
- Medical and dental offices
- Other low-intensity neighborhood-serving uses

These uses are not permitted under the current SR zoning.

Why the Change Is Needed

- SR zoning limits the site to low-density residential and agricultural uses.
 - The Pinal County General Plan allows small-scale commercial uses in this area through a non-major amendment, given the parcel size and intended neighborhood-serving uses.
 - Rezoning enables a community-oriented commercial hub that supports local services and economic activity.
-

Site Plan Summary

- Two buildings, each approximately 10,000 sq. ft.
 - Building heights under 30 feet to remain compatible with nearby homes
 - Parking located between Gantzel Road and the buildings for visibility and easy access
 - Site layout designed to fit the suburban context and maintain a neighborhood-scaled appearance
-

Landscaping & Buffers

- Parking areas screened with low-water-use landscaping
 - Tree buffer along west and south property lines (trees spaced 10 ft on center)
 - Landscaping designed to reduce visual impact and provide noise/privacy separation from adjacent SR-zoned homes
-

Traffic & Access

- Primary access will use the existing curb cut currently serving nearby residential lots
 - A second access point is proposed at the south end of the property to improve circulation and reduce congestion
 - Traffic design aims to minimize neighborhood disruption and maintain safe flow along Gantzel Road
-

Community Benefits

- Convenient access to everyday services close to home
 - New retail, dining, and professional options for the growing area
 - Local job creation and economic activity
 - A small-scale commercial center designed to blend with surrounding development
-

What Is Not Proposed

- No large-scale commercial or industrial uses
 - No buildings over 30 feet
 - No high-intensity or late-night uses
 - No reduction in required landscaping or buffers
-

Questions & Feedback

We welcome your input on the proposed rezoning, site layout, traffic access, and planned uses. Your feedback helps shape a project that fits the needs of the neighborhood.

Neighborhood Meeting Minutes:

Gantzel Commercial Center Rezoning

APN: 104-53-004H

Pinal County, Arizona

Date of Meeting – February 26, 2026

Time of Meeting – 5:00 pm to 9:00 pm

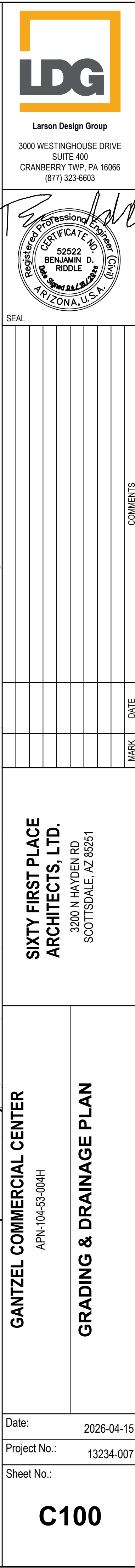
Location of Meeting – QC Library, Zane Grey Room
21802 S. Ellsworth Road
Queen Creek, AZ 85142

Neighborhood Comments:

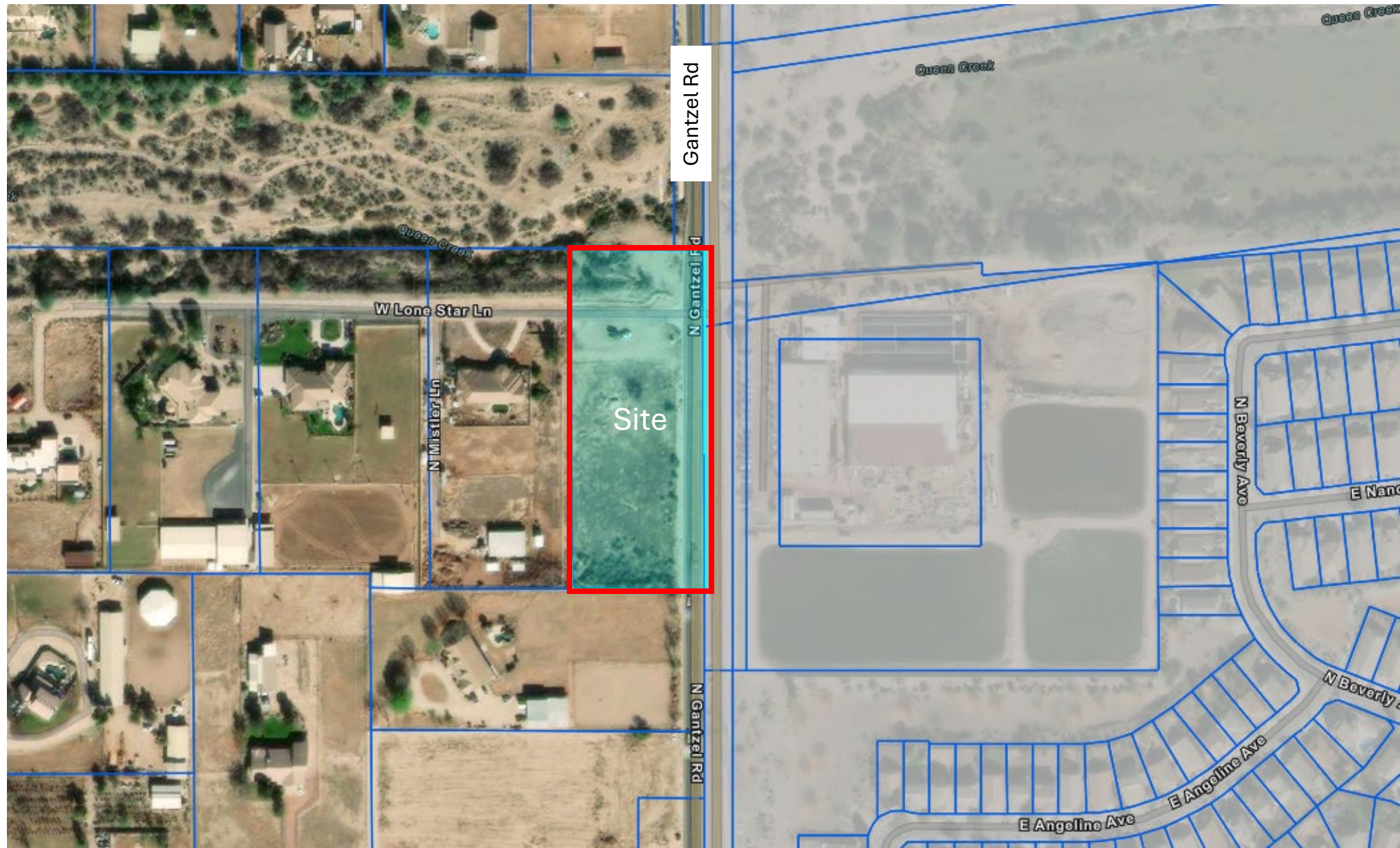
Approximately 8 people attended the meeting. All persons had arrived by 5:15 when introductions were performed and attendees started to voice their concerns about the project. Most speaking by attendees was done by immediate neighbors on Lone Star Lane.

The following items were primary points of concern -

1. Property is located within Sun Valley Farms IV – an HOA.
2. Lone Star Lane is only access to Gantzel Road for neighbors on Lone Star Lane. Concern about crossing commercial land to reach their property. (removed parking along Lone Star Lane). Also concerned about additional traffic exiting onto Gantzel Road
3. Next door neighbor expressed concern about noise and security along their long-shared property line with proposed development.
4. Attendees shared concerns about deliveries and trash pickup in early hours making noise and creating a nuisance.
5. Attendees are afraid of losing their Suburban Ranch feel with commercial development next door.
6. Some attendees expressed concern about new commercial development affecting their property values.
7. Concerns about people lingering behind buildings adjacent to residential properties and possibility of gaining access for crime.



Aerial Map



SWC Gantzel Rd and Lone Star Lane, Pinal County



Gantzel & Lone Star Commercial

Rezone Narrative

Case No. _____

1st Submittal: April 2026



WITHEY
MORRIS
BAUGH

Development Team

Developer:

Lone Star Partners, LLC

2718 East Comstock Street
Gilbert, AZ 85296

Legal Representative:

Withey Morris Baugh PLC

Attorney: Adam Baugh
2121 E. Highland Ave
Phoenix, Arizona 85016

TABLE OF CONTENTS

A. PURPOSE OF REQUEST	4
B. DESCRIPTION OF PROPOSAL	4
C. CONFORMANCE TO THE COMPREHENSIVE PLAN	5
D. LOCATION AND ACCESSIBILITY.....	7
E. UTILITIES AND SERVICES	8
F. NEIGHBORHOOD MEETING SUMMARY.....	9
G. SUPPORTING INFORMATION QUESTIONNAIRE.....	9
H. SUMMARY.....	10

LIST OF EXHIBITS

AERIAL CONTEXT MAP	A
SITE PLAN	B

A. Purpose of Request

On behalf of Lone Star Partners, LLC (the “Applicant”), this application is a request for a rezone from Suburban Ranch (SR) to Community Commercial (C-2) for approximately +/- 3.33 acres located at the southwest corner of Gantzel Road and Lone Star Road (a private access easement), and otherwise commonly known as all or portions of the following Pinal County Assessor’s Parcel Number (APN) 104-53-004H (the “Property”). See **Exhibit A, Aerial Map**.

The purpose of the request is to allow a small-scale neighborhood commercial center on the Property that will have a positive impact on the surrounding area. This development activates an underutilized parcel with an appropriate use adjacent to a major arterial roadway and follows the commercial development and growth pattern along Gantzel Road. The Property has long been vacant, and this development will promote growth in this area.

This proposal provides a diverse mix of retail, service, healthcare, and professional uses. The envisioned development aims to include retail sales, personal and professional services, restaurants, medical offices, dental clinics, and other compatible professional services to serve the growing community.

B. Description of Proposal

i. Proposed Land Uses & Development Standards

The proposed land uses and development standards for the site will comply with those outlined in Chapter 2.320 - Community Commercial (C-2) district.

ii. Nature of the Project

The proposed site plan features commercial buildings designed to be visually accessible and functional within the suburban context. The development is a small retail center composed of two (2) buildings intended for office, retail and restaurant uses. Each multi-tenant building is approximately 9,500 sf each and oriented toward Gantzel Road.

Both structures will be less than 30 feet in height to maintain a neighborhood-friendly scale. Buildings are planned to be positioned back from Gantzel Road, with parking located between the roadway and the structures, ensuring high visibility for signage and ease of access.

Two (2) access points serve the property from Gantzel Road. Access to the site will utilize the existing curb which is a private access easement referred commonly as Lone Star Road. To further improve traffic flow and safety, an additional curb cut is proposed at the south end of the property, reducing reliance on the existing access point and alleviating congestion. Parking will

be carefully screened with low-water-use landscaping to reduce visual impact. 84 parking spaces are provided onsite.

The building is setback 46' feet from the rear property line, and more than 110' feet from the closest residential structure. There is one (1) residential structure and various horse shade structures and corrals behind it. The Site includes a 10' perimeter landscape setback buffer. Buffer zones—comprising trees spaced 10 feet on center—will be established along the west and south property lines, aligning with zoning regulations, and creating a visual and noise barrier from adjacent SR zoning areas.

See Site Plan at **Exhibit B**.

C. Conformance to the Comprehensive Plan

The Property has a Comprehensive Plan land use designation of Rural Living as part of the San Tan Valley Special Area Plan. The Property is currently zoned SR (Suburban Ranch). The Applicant has submitted a minor comprehensive plan amendment application concurrent with this rezone application to change the Property's land use designation within the San Tan Valley Special Area Plan from Rural Living to Suburban Neighborhood. This amendment will allow commercial uses that are more consistent with the recent and growing development pattern along Gantzel Road.

Neighborhood Commercial, which is an anticipated land use under the Suburban Neighborhood designation, is defined "as less than 20 acres and is not shown on the Land Use Plan, but may be included in all land use designations if it addresses the Comprehensive Plan planning guidelines. Neighborhood commercial includes commercial goods and services and typically serves a surrounding residential population."

The Planning Guidelines suggest:

- Neighborhood commercial typically serve a zero to three mile radius market area.
- Neighborhood commercial is located in close proximity to residential to reduce travel time and distance (e.g. to encourage walking, biking and other non-auto related travel modes).
- The size of neighborhood commercial centers should be compatible with surrounding or potential residential uses.
- Neighborhood commercial is best located on collector or arterial streets.
- Neighborhood commercial development should minimize negative impacts of on-site activities to adjacent uses.
- Mitigate negative visual impacts arising from the scale, bulk, and mass of large commercial buildings and centers.
- Promote building designs and construction practices that are sustainable and adapted to Central Arizona's climate.
- Allow for needed flexibility to respond to conditions and constraints inherent to specific sites and areas in Pinal County.

All of these guidelines are demonstrated in the accompanying site plan for this commercial center. In addition, the current development proposal meets many of the goals and strategies outlined in the Pinal County Comprehensive Plan, as outlined below:

3.1 Goal: Carefully manage and guide growth in a manner that promotes economic development, integrates current and future multimodal transportation systems and is sensitive to the natural environment.

The Suburban Neighborhood designation can support a variety of employment-generating businesses and retail services that will promote economic development. In addition, the location of the Property off Gantzel Road, a major transportation corridor, further supports this goal.

3.1.1 Objective: Promote diverse employment opportunities near population base.

The Property is located in the heart of San Tan Valley and will promote a range of employment opportunities. It is expected that the tenants commercial and office tenants in this project will include professional and medical offices, retail uses, personal services, and possibly restaurants – all of which serve residents living in the San Tan area.

4.1 Goal: Improve County and regional mobility.

The Property is located off the Gantzel Road which is a major transportation corridor in San Tan Valley. The new designation will enhance the adjacent roadway to further improve the mobility within Pinal County.

5.1 Goal: Create more employment opportunities in Pinal County.

5.1.1 Objective: Increase the number of businesses that expand in and locate to Pinal County.

5.2 Goal: Encourage the preservation of adequate locations for employment uses to meet long-term economic development needs.

As noted, the Suburban Neighborhood designation can support a variety of employment-generating businesses that will promote economic development. In addition, the location of the Property off Gantzel Road further supports these goals and objectives.

6.1 Goal: Develop a connected system of open space areas that protect and conserve natural, physical and social resources.

6.1.2 Objective: Preserve, protect, or conserve the existing natural drainage system.

The Property is located adjacent to the Queen Creek Wash, which is a significant natural drainage system. The proposal is for development to occur on areas that are not impacted by the drainage systems. This sensitive development also protects natural spaces and resources.

7.1 Goal: Environmental stewardship is part of all decisions made and Pinal County has become a model for sustainability, with environmentally sensitive County codes, policies, and ordinances.

7.1.2.3 Policies: Encourage efforts to integrate areas vegetated with native species into developed areas.

The Proposal will develop a low energy and water use that will incorporate appropriate environmentally sensitive elements in its development. The development will integrate native species into the planned landscaping along the frontage road and western edge.

D. Location and Accessibility

The Property is located entirely within unincorporated Pinal County but within the future planning area of the Town of San Tan Valley, and it is currently undeveloped. The Property fronts Gantzel Road which is a road of regional significance. Gantzel Road is experiencing significant commercial growth. Many properties are changing from rural land into commercial uses to serve the explosive population growth happening in the immediate area.

To the east is Epcor's Pecan Wastewater Treatment Plant. To the north is the Queen Creek dry wash. To the south are more commercial uses along Gantzel Road including Target, Mountainside Fitness, CubeSmart Storage, Dairy Queen, Dutch Bros., RV storage, and a Sprouts-anchored shopping center.)

The proposal is consistent with the Mobility and Connectivity vision component of the Comprehensive Plan. The development will improve Gantzel Road which is a key transportation corridor in San Tan Valley. This further strengthens other existing and planned commercial development to the south of the rezoning site.

The project will be accessed directly from Gantzel Road.

E. Utilities & Services

The Project does not anticipate the need for any improvements of services that would be paid for by the public. The list below identifies the utilities and services applicable to the Property:

Type of Utility / Service	Provider	Contact
Water Services	Arizona Water Company	Arizona Water Company 3805 N. Black Canyon Hwy Phoenix, AZ 85015-5351 602.240.6860 mail@AZwater.com
Wastewater Services	Sewer	Private Sewer
Electrical Services	SRP	SRP Customer Correspondence Customer Communications Services, PAB277 P.O. Box 52025 Phoenix, AZ 85072-2025 bizcenter@srpnet.com (602) 236-0777
Gas Services	Southwest Gas Corporation	Southwest Gas Corporation 1600 E. Northern Avenue Phoenix, AZ 85020-3982
Telecommunications	Cox, Century Link, and Media Com	PO Box 53249 Phoenix, AZ 85072-3249 1-866-272-5777
Police	Pinal County Sheriff's Office	971 N Jason Lopez Circle Building C PO Box 867 Florence, AZ 85132 520-866-5111
Fire	Rural Metro	8465 N. Pima Road Scottsdale, AZ 85258 RuralMetroFire@gmr.net 800.624.5835
Schools	J.O. Combs Unified School District #44	43371 N. Kenworthy Rd. San Tan Valley, Arizona 85140 (480) 987-5300

F. Neighborhood Meeting Summary

A neighborhood meeting was held on February 26, 2026 at the Queen Creek Library. Approximately 8 people attended the meeting. A summary of the neighbor meeting is included with this rezone submittal.

G. Supporting Information Questionnaire (Pinal County)

1. Note any services that are not available to the site. Discuss any improvements of services that would be paid for by the public:
 - a. None
2. What is the amount of traffic to be generated (# of trips/day, deliveries/week)? Show ingress/egress on the site plan:
 - a. On a typical weekday, after full build-out, the Gantzel Commercial Center development is forecasted to generate 1,089 total daily trips, with 79 trips in the morning peak hour and 126 trips in the evening peak hour, according to the traffic study.
3. How many parking spaces are to be provided (employees and customers)?
 - a. 84 parking spaces are to be provided.
4. Is there a potential for excessive noise (I.E.; children, machinery) or the production of smoke, fumes, dust or glare with this proposed land use? If yes, how will you alleviate these problems for your neighbors?
 - a. There is no expectation for excessive noise, odor, dust, gas, glare, etc. It is a simple retail design, and no exterior operations.
5. What type of landscaping are you proposing to screen this use from your neighbors?
 - a. The application proposes a 10' perimeter landscape setback buffer with trees spaced 10 feet on center. This will be planted along the west and south property lines, aligning with zoning regulations.
6. What type of signage are you proposing for the activity?
 - a. Building mounted signage to comply with local codes. Front lit channel letters will be typical signage type.
7. If the proposed land use involves any type of manufacturing or production process, provide a short synopsis of the processes utilizing diagrams, flowcharts and/or a short narrative:
 - a. Not applicable
8. Explain how the appearance and operation of the proposed land use will maintain the integrity and character of the zone in which the use is requested
 - a. The proposal creates opportunities for employment and commercial uses to help balance the surrounding uses in a meaningful way and responds to the population growth that is occurring in the San Tan area. Commercial uses are encouraged and expected along this road of regional significance

and provide a new injection of economic support desired in the new Town of San Tan Valley.

9. Have you discussed possible conditions that may be placed on the approval with the Planning Department?

a. None.

10. Do you understand that if a condition is violated, that there is a public process by which your zoning may be reverted?

a. Yes

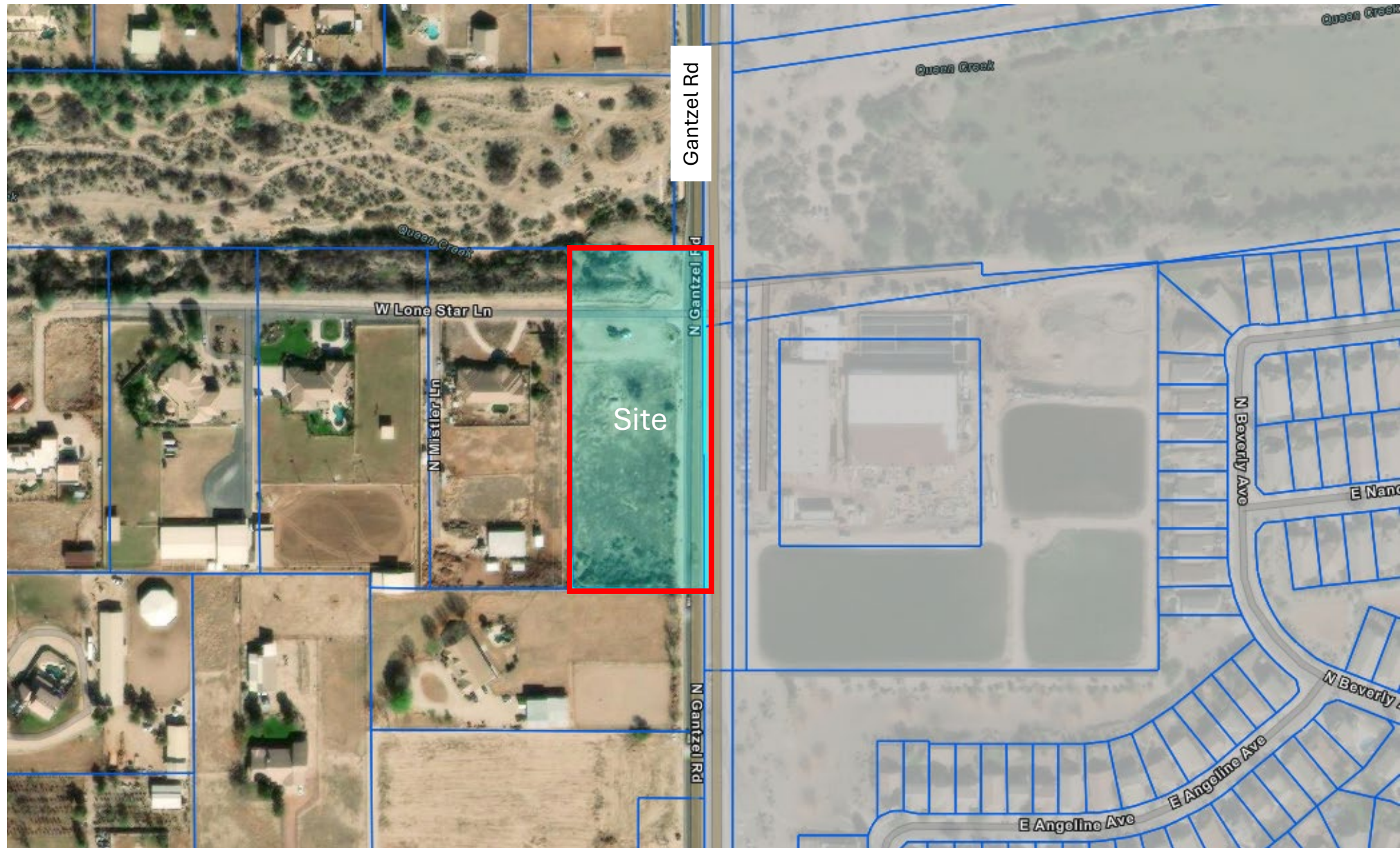
H. Summary

As the newly incorporated San Tan Valley Town commences, the need for commercial uses and sales tax revenue cannot be understated. The proposal creates opportunities for employment and commercial uses to help balance the surrounding uses in a meaningful way and responds to the population growth that is occurring in the San Tan area. Commercial uses are encouraged and expected along this road of regional significance and provide a new injection of economic support desired in the new Town of San Tan Valley.

Commercial zoning can support a variety of employment-generating businesses and retail services that will promote economic development. In addition, the location of the Property off Gantzel Road, a major transportation corridor, further supports this zoning change. The new designation will strengthen other existing commercial development in the area, will help grow San Tan Valley, and promote economic development in this rapidly growing area.

EXHIBIT A

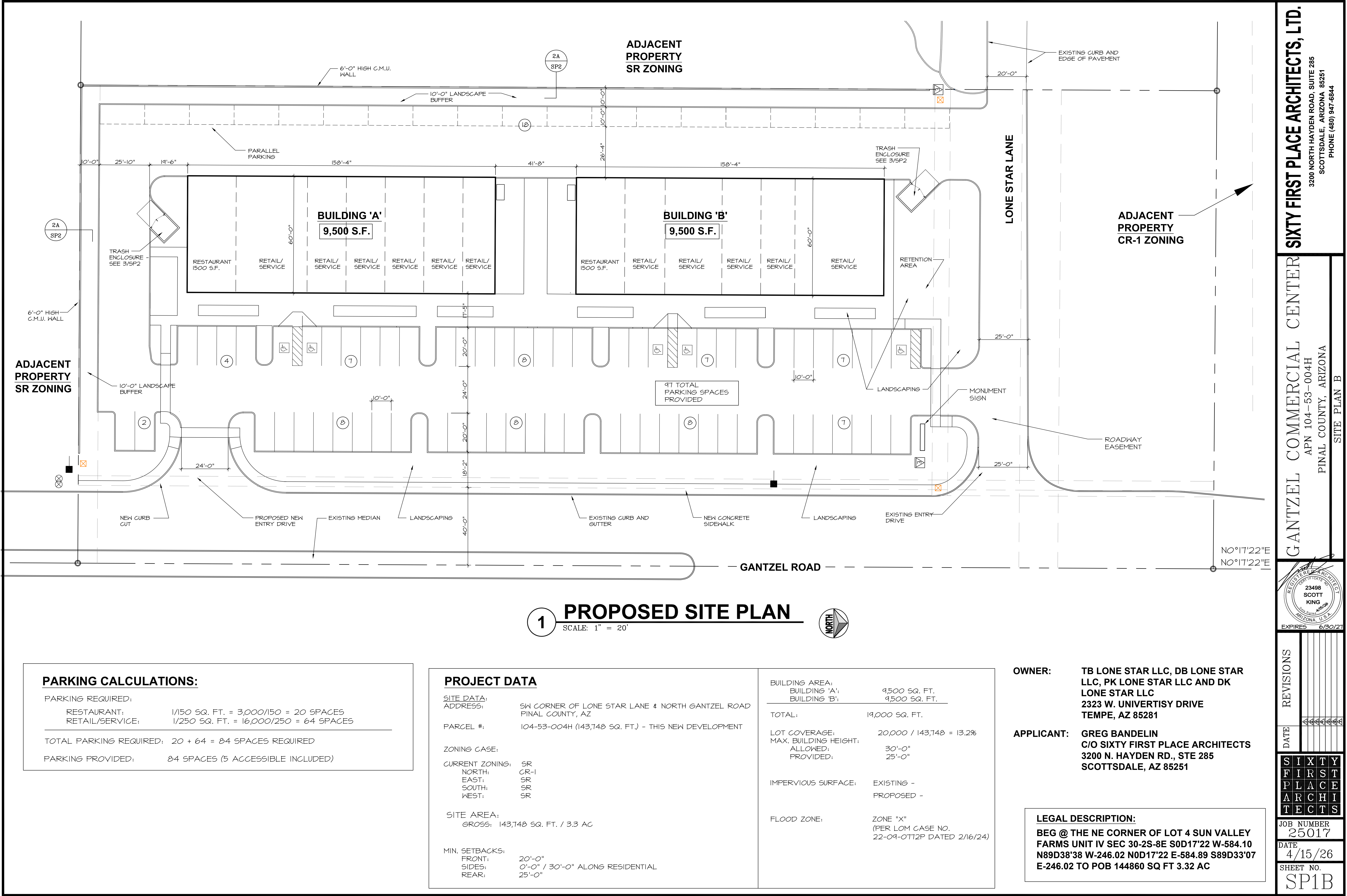
Aerial Map



SWC Gantzel Rd and Lone Star Lane, Pinal County



EXHIBIT B



PARKING CALCULATIONS:

PARKING REQUIRED:

RESTAURANT:	1/150 SQ. FT. = 3,000/150 = 20 SPACES
RETAIL/SERVICE:	1/250 SQ. FT. = 16,000/250 = 64 SPACES

TOTAL PARKING REQUIRED: 20 + 64 = 84 SPACES REQUIRED

PARKING PROVIDED: 84 SPACES (5 ACCESSIBLE INCLUDED)

PROJECT DATA

SITE DATA:
ADDRESS: SW CORNER OF LONE STAR LANE & NORTH GANTZEL ROAD
PINAL COUNTY, AZ

PARCEL #: 104-53-004H (143,748 SQ. FT.) - THIS NEW DEVELOPMENT

ZONING CASE:

CURRENT ZONING: SR
NORTH: CR-1
EAST: SR
SOUTH: SR
WEST: SR

SITE AREA:
GROSS: 143,748 SQ. FT. / 3.3 AC

MIN. SETBACKS:
FRONT: 20'-0"
SIDES: 0'-0" / 30'-0" ALONG RESIDENTIAL
REAR: 25'-0"

BUILDING AREA:
BUILDING 'A': 9,500 SQ. FT.
BUILDING 'B': 9,500 SQ. FT.

TOTAL: 19,000 SQ. FT.

LOT COVERAGE: 20,000 / 143,748 = 13.2%

MAX. BUILDING HEIGHT:
ALLOWED: 30'-0"
PROVIDED: 25'-0"

IMPERVIOUS SURFACE: EXISTING -
PROPOSED -

FLOOD ZONE: ZONE "X"
(PER LOM CASE NO. 22-09-07T2P DATED 2/16/24)

OWNER: TB LONE STAR LLC, DB LONE STAR LLC, PK LONE STAR LLC AND DK LONE STAR LLC
2323 W. UNIVERTISY DRIVE
TEMPE, AZ 85281

APPLICANT: GREG BANDELIN
C/O SIXTY FIRST PLACE ARCHITECTS
3200 N. HAYDEN RD., STE 285
SCOTTSDALE, AZ 85251

LEGAL DESCRIPTION:
BEG @ THE NE CORNER OF LOT 4 SUN VALLEY FARMS UNIT IV SEC 30-2S-8E S0D17'22 W-584.10 N89D38'38 W-246.02 N0D17'22 E-584.89 S89D33'07 E-246.02 TO POB 144860 SQ FT 3.32 AC

SIXTY FIRST PLACE ARCHITECTS

3200 NORTH HAYDEN ROAD, SUITE 285
SCOTTSDALE, ARIZONA 85251
PHONE (480) 947-6844

GANTZEL COMMERCIAL CENTER

APN 104-53-004H
PINAL COUNTY, ARIZONA

SITE PLAN B

REGISTERED ARCHITECT
23498
SCOTT KING
ARIZONA, U.S.A.
EXPIRES 6/30/27

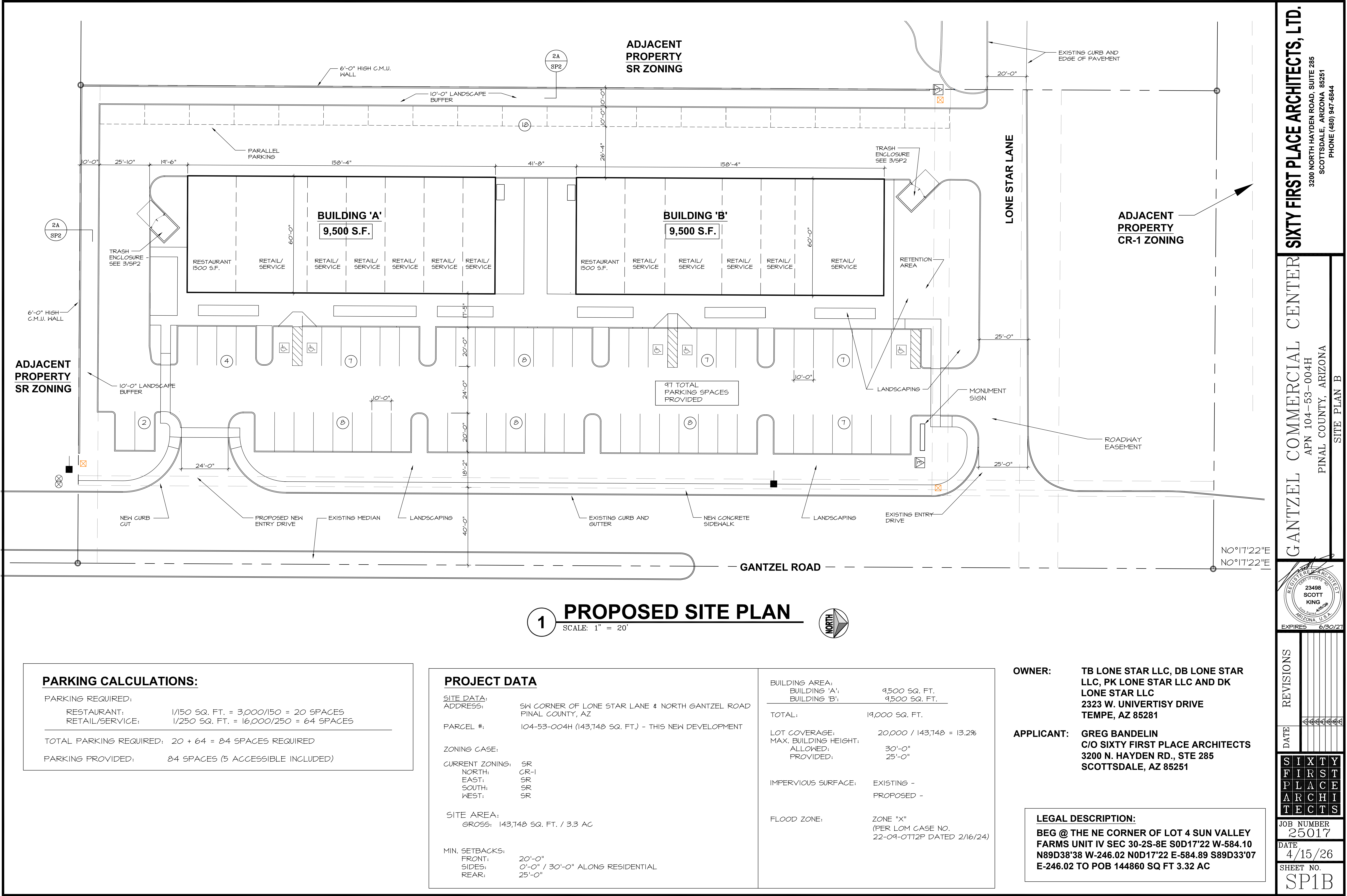
REVISIONS

DATE	REVISIONS

JOB NUMBER
25017

DATE
4/15/26

SHEET NO.
SP1B



PARKING CALCULATIONS:

PARKING REQUIRED:

RESTAURANT:	1/150 SQ. FT. = 3,000/150 = 20 SPACES
RETAIL/SERVICE:	1/250 SQ. FT. = 16,000/250 = 64 SPACES

TOTAL PARKING REQUIRED: 20 + 64 = 84 SPACES REQUIRED

PARKING PROVIDED: 84 SPACES (5 ACCESSIBLE INCLUDED)

PROJECT DATA

SITE DATA:
ADDRESS: SW CORNER OF LONE STAR LANE & NORTH GANTZEL ROAD
PINAL COUNTY, AZ

PARCEL #: 104-53-004H (143,748 SQ. FT.) - THIS NEW DEVELOPMENT

ZONING CASE:

CURRENT ZONING: SR
NORTH: CR-1
EAST: SR
SOUTH: SR
WEST: SR

SITE AREA:
GROSS: 143,748 SQ. FT. / 3.3 AC

MIN. SETBACKS:
FRONT: 20'-0"
SIDES: 0'-0" / 30'-0" ALONG RESIDENTIAL
REAR: 25'-0"

BUILDING AREA:
BUILDING 'A': 9,500 SQ. FT.
BUILDING 'B': 9,500 SQ. FT.

TOTAL: 19,000 SQ. FT.

LOT COVERAGE: 20,000 / 143,748 = 13.2%

MAX. BUILDING HEIGHT:
ALLOWED: 30'-0"
PROVIDED: 25'-0"

IMPERVIOUS SURFACE: EXISTING -
PROPOSED -

FLOOD ZONE: ZONE "X"
(PER LOM CASE NO. 22-09-07T2P DATED 2/16/24)

OWNER: TB LONE STAR LLC, DB LONE STAR LLC, PK LONE STAR LLC AND DK LONE STAR LLC
2323 W. UNIVERTISY DRIVE
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SIXTY FIRST PLACE ARCHITECTS

3200 NORTH HAYDEN ROAD, SUITE 285
SCOTTSDALE, ARIZONA 85251
PHONE (480) 947-6844

GANTZEL COMMERCIAL CENTER

APN 104-53-004H
PINAL COUNTY, ARIZONA

REVISIONS

DATE	REVISIONS

JOB NUMBER
25017

DATE
4/15/26

SHEET NO.
SP1B

TRAFFIC IMPACT ANALYSIS

Gantzel Commercial Center

March 5, 2026

PREPARED FOR
Sixty First Place Architects
3200 North Hayden Road, Suite 285
Scottsdale, Arizona 85281

PREPARED BY



APPROVED BY:

PINAL COUNTY ENGINEER
PINAL COUNTY DEPARTMENT OF PUBLIC WORKS

DATE

APPROVAL EXPIRES:

DATE

County Case Numbers: Z-PA-128-25

1st Review by xx x/xx/xxxx

TRAFFIC IMPACT ANALYSIS

Gantzel Commercial Center

March 5, 2026

UCG Project Number: TR26013

PREPARED FOR
Sixty First Place Architects
3200 North Hayden Road, Suite 285
Scottsdale, Arizona 85281

PREPARED BY



United Civil Group

2803 N. 7th Avenue
Phoenix, Arizona 85007
602-265-6155



CONDUCTED BY

David Kinnear, PE
Project Manager

TABLE OF CONTENTS

I. INTRODUCTION AND SUMMARY	1
A. PURPOSE OF THE REPORT	1
B. GUIDING STANDARDS	1
C. STUDY OBJECTIVES.....	1
D. CONCLUSIONS AND RECOMMENDATIONS	2
II. PROPOSED DEVELOPMENT.....	3
A. SITE LOCATION	3
B. LAND USE.....	3
C. PHASING AND TIMING.....	3
D. SITE ACCESSIBILITY.....	3
III. STUDY AREA CONDITIONS.....	7
A. STUDY AREA.....	7
B. STUDY AREA LAND USE.....	7
C. ANTICIPATED FUTURE DEVELOPMENT AND PLANNED IMPROVEMENTS	7
IV. EXISTING ROADWAY CONDITIONS	8
A. PHYSICAL CHARACTERISTICS	8
B. EXISTING TRAFFIC VOLUMES.....	9
C. EXISTING TRAFFIC OBSERVATIONS AND ISSUES	9
D. CRASH DATA	11
E. INTERSECTION LEVEL OF SERVICE ANALYSES.....	13
E.1 HCM Capacity Analyses and Levels of Service	13
E.2 Existing Intersection Level of Service.....	14
V. PROJECTED TRAFFIC	17
A. TRIP GENERATION	17
B. TRIP DISTRIBUTION	17
C. PROJECTED BACKGROUND TRAFFIC	20
VI. TRAFFIC AND IMPROVEMENT ANALYSIS	24
A. ACCESSIBILITY TO GANTZEL COMMERCIAL CENTER DEVELOPMENT	24
B. AUXILIARY LANE ANALYSES	24
B.1. Right Turn Lane Warrants	24
B.2. Right Lane Queue and Storage Length Analysis	25
C. TRAFFIC SIGNAL WARRANT ANALYSIS	25
D. INTERSECTION SIGHT DISTANCE.....	27
E. TOTAL TRAFFIC.....	27
F. ROADWAY AND INTERSECTION GEOMETRIC IMPROVEMENTS	31
VII. CONCLUSIONS AND RECOMMENDATIONS	33
VIII. LIMITATIONS.....	34
IX. SOURCES	35

LIST OF TABLES

TABLE 1: INJURY SEVERITY.....	11
TABLE 2: COLLISION MANNER	12
TABLE 3: INTERSECTION LEVELS OF SERVICE AND DELAY	13
TABLE 4: 2026 EXISTING CONDITIONS INTERSECTION LEVELS OF SERVICE	15
TABLE 5: PROPOSED DEVELOPMENT TRIP GENERATION.....	17
TABLE 6: TRIP DISTRIBUTION PERCENTAGES	18
TABLE 7: 2027 BACKGROUND CONDITIONS INTERSECTION LEVELS OF SERVICE.....	22
TABLE 8: RIGHT TURN LANE ANALYSIS.....	25
TABLE 9: 2027 TOTAL CONDITIONS INTERSECTION LEVELS OF SERVICE	30

LIST OF FIGURES

FIGURE 1: VICINITY MAP.....	4
FIGURE 2: AERIAL VIEW	5
FIGURE 3: SITE PLAN.....	6
FIGURE 4: EXISTING CONDITIONS AND TRAFFIC VOLUMES – YEAR 2026	10
FIGURE 5: SITE GENERATED TRAFFIC AND TRIP DISTRIBUTION	19
FIGURE 6: BACKGROUND TRAFFIC – YEARS 2027.....	21
FIGURE 7: TOTAL TRAFFIC – YEAR 2027	28
FIGURE 8: SITE TRIANGLE	29
FIGURE 9: RECOMMENDATIONS.....	32

APPENDIX A . . . Traffic Counts
APPENDIX B . . . Capacity Analyses

I. INTRODUCTION AND SUMMARY

A. Purpose of the Report

Sixty First Place Architects retained United Civil Group to perform this Traffic Impact Analysis (TIA) for the proposed Gantzel Commercial Center Development. The Gantzel Commercial Center is located on the southwest corner of Gantzel Road and Lone Star Lane in Pinal County, Arizona. The approximate 3.3-acre development is planned to include two 10,000 square foot commercial buildings, which will include restaurant and retail spaces. The development will be constructed in one phase with full buildout of Gantzel Commercial Center anticipated by 2027.

B. Guiding Standards

United Civil Group performed this TIA in general accordance with the requirements as specified in the Pinal County *Traffic Impact Assessment Guidelines & Procedures* dated January 2007, locally accepted standards, and industry practice.

Per the Pinal County Guidelines & Procedures, this development is characterized as a small development with less than 500 peak hour trips. Therefore, the study horizon will be the opening year (year 2027). The minimum study area limits include all site access driveways and signalized intersections and/or major unsignalized street intersections within 1/2 mile of the site boundary.

The study area intersections are defined within 1/2 mile of the site and include the intersections of Gantzel Road/Lone Star Lane, Gantzel Road/Chandler Heights Road, and Gantzel Road/Pecan Creek Drive.

C. Study Objectives

This study is intended to investigate the existing and future traffic conditions and identify any potential roadway improvements necessary to serve the Gantzel Commercial Center Development. These recommended improvements may include warrant studies and installation of a new traffic signal, turn lanes and other offsite paving improvements to mitigate traffic. Major study objectives of this traffic report are as follows:

- Determine the existing morning and evening peak hour traffic volumes at the study area intersections.
- Analyze the study area intersections as well as the planned site access intersections for the development through the horizon years.

- Where applicable, recommend safety, intersection and/or roadway improvements sufficient to meet the needs of the development and adjacent roadway network due to the additional site generated traffic volumes.

D. Conclusions and Recommendations

The approximate 3.3-acre development is planned to include two 10,000 square foot commercial buildings, which will include restaurant and retail spaces. The development will be constructed in one phase with full buildout of Gantzel Commercial Center anticipated by 2027.

The Gantzel Commercial Center Development will have three accesses. Access A is located on Gantzel Road, Access B and C is located on Lone Star Lane. Gantzel Road and Lone Star Lane are owned and maintained by Pinal County. Therefore, Accesses A, B and C were analyzed using Pinal County Traffic Engineering Guidelines.

On a typical weekday, after full build-out, the Gantzel Commercial Center development is forecasted to generate 1,089 total daily trips, with 79 trips in the morning peak hour and 126 trips in the evening peak hour, according to the 11th Edition of the Trip Generation Manual.

A right-turn lane is warranted on Gantzel Road at Access A, with a 100-foot storage and 120-foot taper.

Traffic signals are not warranted based on traffic generated by the surrounding developments including the Gantzel Commercial Center Development. Signal warrants should be conducted to confirm signals are warranted prior to their installation.

Proper intersection sight-distance and sight triangles shall be provided and maintained at all site access driveways of the proposed development to give drivers exiting the site a clear view of oncoming traffic. The landscape and hardscape (monument signs) within the sight triangles must not obstruct the driver's view of the adjacent travel lanes. The sight triangles shall be provided on the developer's landscape and design plans.

II. PROPOSED DEVELOPMENT

A. Site Location

The Gantzel Commercial Center Development is planned for the undeveloped parcel of land located on the southwest corner of Gantzel Road/Lone Star Lane in Pinal County, Arizona. **Figure 1 and Figure 2** present the location of the proposed Gantzel Commercial Center Development within the context of the immediate area and its location within Pinal County. **Figure 3** presents the site plan layout for the proposed Gantzel Commercial Center Development.

B. Land Use

The approximate 3.3-acre development is currently agricultural land. In the future, Gantzel Commercial Center is planned to include two 10,000 square foot commercial buildings, which will include restaurant and retail spaces.

C. Phasing and Timing

The Gantzel Commercial Center Development is planned to be developed in one phase, year 2026.

D. Site Accessibility

The Gantzel Commercial Center Development will have three accesses. Access A is located on Gantzel Road, Accesses B and C are located on Lone Star Lane. Gantzel Road and Lone Star Lane are owned and maintained by Pinal County. Therefore, Accesses A, B and C were analyzed using Pinal County Traffic Engineering Guidelines.

- **Access A** is located on Gantzel Road and will be constructed as a right-in-right-out driveway. Access A is located approximately 430 feet south of Lone Star Lane.
- **Access B** is located on Lone Star Lane and will be constructed as a full access driveway. Access B is located approximately 53 feet west of Gantzel Road.
- **Access C** is located on Lone Star Lane and will be constructed as a full access driveway, which will primarily be used for delivery drop-off. Access C is located approximately 175 feet west of Gantzel Road.

Figure 1: Vicinity Map

Figure 2: Aerial View

Figure 3: Site Plan

III. STUDY AREA CONDITIONS

A. Study Area

Based on the forecasted trip generation of the Gantzel Commercial Center development (see Section V. Projected Traffic, Table 5), the proposed development falls under a Category I – Small Development per the County's *Traffic Impact Assessment Guidelines & Procedures, January 2007*. According to these guidelines, the study area includes the intersections of:

- Gantzel Road/Lone Star Lane
- Gantzel Road/Chandler Heights Road
- Gantzel Road/Pecan Creek Drive

B. Study Area Land Use

The following describes the existing land uses of the subject site and surrounding area:

SUBJECT SITE: Undeveloped agricultural land

NORTH: Undeveloped agricultural land, followed by residential homes

SOUTH: Residential homes

EAST: Gantzel Road followed by Pecan Creek Wastewater Treatment Plant

WEST: Residential homes followed by undeveloped agricultural land

C. Anticipated Future Development and Planned Improvements

There are no known developments that are planned within the vicinity of Gantzel Commercial Center Development at this time.

IV. EXISTING ROADWAY CONDITIONS

A. Physical Characteristics

Gantzel Road, located on the mile section, is classified as a regionally significant principal arterial according to the Pinal County Development Services website accessed February 25, 2026. Adjacent to the subject site, Gantzel Road is constructed as a four-lane section with two lanes in the northbound and southbound directions, separated by an approximate 14-foot median. The posted speed limit on Gantzel Road is 45 mph.

Lone Star Lane is classified as a local roadway. Adjacent to the subject site, Lone Star Lane is paved with one-lane in each direction. There is no posted speed limit on Lone Star Lane.

Chandler Heights Road, located on the mile section, is classified as a east-west collector in accordance with the Pinal County Development Services website accessed February 25, 2026. Chandler Heights Road is paved with one lane in each direction. The posted speed limit is 35mph.

Pecan Creek Drive, located on the mile section, is classified as a east-west minor arterial in accordance with the Pinal County Development Services website accessed February 25, 2026. Pecan Creek Drive is paved with one lane in each direction with a two-way-left-turn lane. The posted speed limit is 30mph.

The intersection of **Gantzel Road/Lone Star Lane** operates as an unsignalized intersection with the stop condition on the eastbound approach. The northbound approach consists of a 120' two-way-left-turn lane, two through lanes and a bike lane. The southbound approach consists of an exclusive right-turn lane (75 feet of storage and a 60-foot taper), two through lanes and a bike lane. The eastbound approach consists of one shared left-right turn lane.

The intersection of **Gantzel Road/Chandler Heights Road** operates as a signalized intersection. The northbound approach consists of an exclusive left-turn lane (110 feet of storage and a 60-foot gap), two through lanes and a dedicated right-turn lane (110 feet of storage and a 180-foot taper). The southbound approach consists of an exclusive left-turn lane (110 feet of storage and a 180-foot taper), two through lanes and a dedicated right-turn lane (110 feet of storage and a 180-foot taper). The eastbound/westbound approaches consist of one left-turn lane and one shared thru-right turn lane.

The intersection of **Gantzel Road/Pecan Creek Drive** operates as a signalized intersection. The northbound approach consists of an exclusive left-turn lane (200 feet of storage and a 90-foot gap), two through lanes and a dedicated right-turn lane (190 feet of storage and a 165-foot taper). The southbound approach consists of an exclusive

left-turn lane (110 feet of storage and a 90-foot taper), two through lanes and a dedicated right-turn lane (70 feet of storage and an 80-foot taper). The eastbound/westbound approaches consist of one left-turn lane and one shared through-right turn lane.

B. Existing Traffic Volumes

Existing turning movement counts in 15-minute intervals were collected at the study area intersections of Gantzel Road/Lone Star Lane, Gantzel Road/Chandler Heights Road, and Gantzel Road/Pecan Creek Drive on Wednesday, February 11, 2026, during the morning peak period (7:00AM – 9:00AM) and evening peak period (4:00PM – 6:00PM).

MAG's Data Analytics Highway Performance Monitoring (HPMS) website was utilized to provide average daily traffic volumes for the roadway within the vicinity of the site. The roadway within the study area that are currently included in HPMS are the paved roadway of Gantzel Road. The traffic volumes for Gantzel Road near the subject site is 18,318 and 18,260 in the northbound and southbound directions, respectively.

The resulting morning and evening peak hour traffic volumes at the study area intersections along with the ADTs are presented in **Figure 4**. Complete traffic volume data can be found in *Appendix A: Traffic Counts*.

Figure 4 also graphically depicts the existing intersection geometry and the existing traffic volumes at the study area intersections.

C. Existing Traffic Observations and Issues

Traffic conditions and operations were observed at the study area intersections during the morning and evening peak periods, and no traffic issues were noted.

Figure 4: Existing Conditions and Traffic Volumes – Year 2026

D. Crash Data

Three years of the most recently provided crash data (2023 – 2025) were obtained from the Arizona Department of Transportation (ADOT) Arizona Crash Information System (ACIS). The data was queried for the intersections of:

- Gantzel Road/Lone Star Lane
- Gantzel Road/Chandler Heights Road
- Gantzel Road/Pecan Creek Drive

The crash data summarized in **Tables 1 and 2** for the study intersections are crashes that were reported to law enforcement. Table 1 summarizes the total number of crashes at the intersection by severity. Crash severity is determined by the reporting officer at the time of the crash or soon thereafter based on the most severe injury sustained by an involved party. Crashes are shown from most severe (Fatal) to least severe (No Injury). Table 2 summarizes the total number of crashes at the intersection by collision type. Crash type is designated by the reporting officer as the first injury or damage producing event that characterizes the crash type and identifies the nature of the crash. The categories within this report include Single Vehicle, Rear End, Left Turn, Angle (front to side, other than left turn), and Other/unknown.

An offset of 250 feet was used on all legs of each intersection over the three-year study period.

Table 1: Injury Severity

Year	Fatal	Suspected Serious Injury	Suspected Minor Injury	Possible Injury	No Injury	Total
Gantzel Road/Lone Star Lane						
2023			2		2	4
2024						0
2025			1		2	2
Total	0	0	2	0	4	6
Gantzel Road/Chandler Heights Road						
2023					12	12
2024			2	4	4	10
2025					6	6
Total	0	0	2	4	22	28
Gantzel Road/Pecan Creek Drive						
2023		1		6	27	34
2024			3	2	17	22
2025		2	2	4	10	18
Total	0	3	5	12	54	74

Table 2: Collision Manner

Year	Single Vehicle	Angle	Left Turn	Rear End	Head On	Sideswipe Same Direction	Sideswipe Opposite Direction	Rear to Side	Unknown /Other	Total
Gantzel Road/Lone Star Lane										
2023				4						4
2024										0
2025				2						2
Total	0	0	0	6	0	0	0	0	0	6
Gantzel Road/Chandler Heights Road										
2023		6	2	4						12
2024		2	2	2		2			2	10
2025		2	2	2						6
Total	0	10	6	8	0	2	0	0	2	28
Gantzel Road/Pecan Creek Drive										
2023	1	4	8	17		4				34
2024		2	8	10		2				22
2025		2	4	8		4				18
Total	1	8	20	35	0	10	0	0	0	74

Based on the data provided in ACIS, it is difficult to determine if crash patterns exist at the study area intersection due to the variability in the data. To determine if crash patterns exist, individual crash records would be required, and a safety analysis performed.

E. Intersection Level of Service Analyses

E.1 HCM Capacity Analyses and Levels of Service

Intersection capacity analysis is a principal tool used in traffic engineering. Operation is characterized according to the amount of delay at an intersection approach and quantified into a level of service (LOS). The intersection LOS was determined using the methodologies presented in the Transportation Research Board's Highway Capacity Manual (HCM). The LOS grades quantify and categorize a driver's discomfort, frustration, fuel consumption, and travel times experienced because of intersection control and the resulting traffic queuing. Per the HCM, the signalized and unsignalized (all-way stop controlled or two-way stop-controlled intersection) delay and associated LOS is presented in **Table 3**. Pinal County strives to obtain a level of service C or better for both signalized and unsignalized intersection overall operations. Intersections having a LOS D, E, or F may warrant improvements or traffic reductions.

Table 3: Intersection Levels of Service and Delay

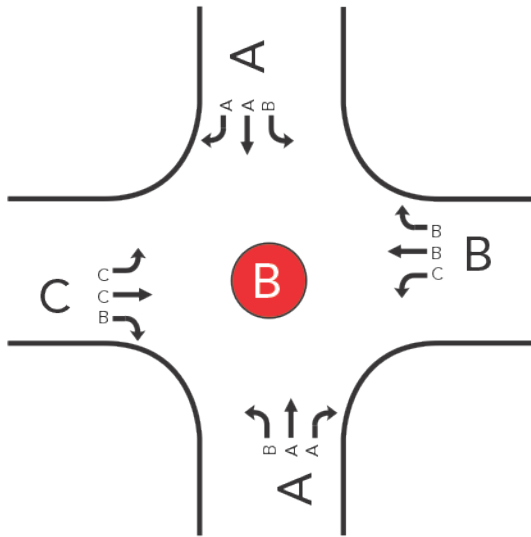
Level of Service	Description	Signalized Delay (Sec/Veh)	Unsignalized Delay (Sec/Veh)
A	Minimal control delay, traffic operates at primary free flow conditions, unimpeded movement within traffic stream	≤ 10	≤ 10
B	Minor control delay at signalized intersections, traffic operates at a fairly unimpeded level with slightly restricted movement within traffic stream	> 10 and ≤ 20	> 10 and ≤ 15
C	Moderate control delay, movement within traffic stream more restricted than LOS B, formation of queues contributes to lower average travel speeds	> 20 and ≤ 35	> 15 and ≤ 25
D	Considerable control delay that may be substantially increased by small increases in flow, average travel speeds continue to decrease.	> 35 and ≤ 55	> 25 and ≤ 35
E	High control delay, average travel speed no more than 22 percent of free flow speed	> 55 and ≤ 80	> 35 and ≤ 50
F	Extremely high control	> 80	> 50

Source: Highway Capacity Manual 2010

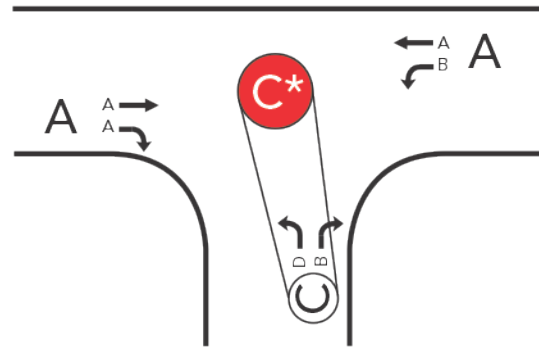
For signalized and all-way stop controlled intersections, LOS is calculated for a movement (e.g., left, through, right), for the approach (e.g., northbound, southbound, eastbound, westbound) and for the overall intersection.

EXAMPLE:

Signalized & All-Way Stop Controlled



Two-Way Stop Controlled



*Reported as approach LOS

Source: United Civil Group, 2019

E.2 Existing Intersection Level of Service

The level of service (LOS) and average delay at the existing study area intersections were evaluated using the 2026 intersection volumes, lane geometry, and existing traffic control as presented in Figure 4. PTV Vistro traffic modeling software, employing the methodologies as presented in the Highway Capacity Manual (HCM), was utilized for the capacity analyses to obtain the existing conditions levels of service. Summaries of the Vistro output calculations are included in *Appendix B: Capacity Analyses*.

The results of the existing levels of service analysis are presented in **Table 4**.

Table 4: 2026 Existing Conditions Intersection Levels of Service

Intersection and Approach		Turning Movement	Delay (s/veh) for Movement & LOS AM (PM)	Approach Delay s/veh & LOS AM (PM)	Intersection Delay s/veh & LOS AM (PM)*
Gantzel Rd & Chandler Heights Rd – Signalized	NB	Left	9.1A (14.09B)	22.14C (21.02C)	20.92C (23.55C)
		Thru	22.98C (21.84C)		
		Right	11.51B (15.39B)		
	SB	Left	14.59B (14.37B)	16.16B (25.31C)	
		Thru	16.23B (26.24C)		
		Right	10.77B (12.29B)		
	EB	Left	30.24C (24.43C)	23.24C (22.54C)	
		Thru	23.1C (22.51C)		
		Right	23.1C (22.51C)		
	WB	Left	30.82C (29.46C)	29.03C (26.13C)	
		Thru	27.56C (22.23C)		
		Right	27.56C (22.23C)		
Gantzel Rd & Lone Star Ln – Two-way stop	NB	Left	11.17B (14.2B)	0A (0A)	67.41F (131.47F)
		Thru	0A (0A)		
		Right	0A (0A)		
	SB	Left	14.23B (11.8B)	0.04A (0A)	
		Thru	0A (0A)		
		Right	0A (0A)		
	EB	Left	27.53D (42.65E)	22.79C (42.65E)	
		Thru	34.26D (35.19E)		
		Right	13.29B (16.69C)		
	WB	Left	163.56F (130.43F)	67.41F (131.47F)	
		Thru	217.23F (250.24F)		
		Right	19.33C (13.74B)		
Gantzel Rd & Pecan Creek Dr – Signalized	NB	Left	16.02B (13.78B)	32.33C (18.37B)	29.36C (21.9C)
		Thru	34.18C (19.21B)		
		Right	18.48B (13.99B)		
	SB	Left	22.31C (27.65C)	25.96C (22.74C)	
		Thru	26.83C (22.03C)		
		Right	17.74B (12.85B)		
	EB	Left	17.86B (21.68C)	20.21C (24.98C)	
		Thru	21.69C (28.04C)		
		Right	21.69C (28.04C)		
	WB	Left	17.79B (21.83C)	30.49C (31.27C)	
		Thru	36.72D (35.81D)		
		Right	36.72D (35.81D)		

As shown, the study area intersections of Gantzel Road/Chandler Heights Road and Gantzel Road/Pecan Creek Drive currently operate at acceptable levels of service, LOS C or better, in the morning and evening peak hours. However, the intersection of Gantzel Road/Lone Star Lane operates at an unacceptable level of service, LOS F. This is because motorists wanting to make a left turn from the stop-controlled leg experience delay. Left-turning and through movements on stop-controlled minor roads that intersect with major streets typically experience greater delay for short periods of time in the peak hours due to the wait for acceptable gaps on the major street, while the free-flowing major streets experience minimal delay.

V. PROJECTED TRAFFIC

A. Trip Generation

Estimates of the traffic volumes that will be generated by the development were determined from transportation planning data taken from the Institute of Transportation Engineers (ITE) *Trip Generation Manual, 12th Edition, 2025*. The ITE rates are based on studies that measure trip generation characteristics for various types of land uses. The rates are expressed in terms of trips per unit of land use type.

LUC 822 – Strip Retail Plaza: A strip retail plaza is an integrated group of commercial establishments planned, developed, owned, and managed as a unit. Each study site in this land use has less than 40,000 square feet of gross leasable area (GLA). Because a strip retail plaza is open-air, its GLA is the same as the gross floor area of the building.

Table 5: Proposed Development Trip Generation

Land Use	ITE Code	Units	Total Size	Daily	AM Peak			PM Peak		
					Total	In	Out	Total	In	Out
Strip Retail Plaza	822	Sq.Ft.	20	1,089	79	43	36	126	63	63

Strip Retail Plaza – ITE LUC 822

Daily	$T = 7.81 \times (1000 \text{ Sq.Ft.})$	50% entering, 50% exiting
AM Peak Hour	$T = 3.93 \times (1000 \text{ Sq.Ft.})$	55% entering, 45% exiting
PM Peak Hour	$T = 6.29 \times (1000 \text{ Sq.Ft.})$	50% entering, 50% exiting

On a typical weekday, the proposed development is estimated to generate an increase of 1,089 trips daily with 79 additional trips in the morning peak hour and an additional 126 trips in the evening peak hour.

B. Trip Distribution

The trip distribution procedure determines the general pattern of travel for vehicles entering and leaving the subject site and the study area. The assumed trip distribution percentages for the development are shown in **Table 6**. For a development of this type, these percentages are mainly based on the type of land uses of the development, the location of the site within Pinal County, and the connectivity of the site to the region.

Table 6: Trip Distribution Percentages

Direction	Trip Distribution Percentage	
	Arriving From	Departing To
Gantzel Road north of Chandler Heights Road	45%	45%
Gantzel Road south of Pecan Creek Drive	40%	40%
Chandler Heights Road east of Gantzel Road	8%	8%
Chandler Heights Road west of Gantzel Road	2%	2%
Pecan Creek Road east of Gantzel Road	5%	5%
Pecan Creek Road west of Gantzel Road	0%	0%

Figure 5 presents the site generated traffic to and from Gantzel Commercial Center proposed development.

Figure 5: Site Generated Traffic and Trip Distribution

C. Projected Background Traffic

Non-site or background traffic volumes representing the amount of traffic estimated to be on the area roadway network without the proposed development within the study area were projected for the horizon year of the development, year 2027. A yearly ambient growth rate for the area of two percent was assumed for the horizon year.

Capacity analyses at the existing study area intersections were performed for the forecasted background traffic for the horizon year of the study, as presented in **Figure 6**.

Figure 6: Background Traffic – Years 2027

Table 7 below presents the background levels of service at the study area intersection without the proposed development and without any of the recommended improvements. Complete capacity analyses are provided in *Appendix B: Capacity Analyses*.

Table 7: 2027 Background Conditions Intersection Levels of Service

Intersection and Approach		Turning Movement	Delay (s/veh) for Movement & LOS AM (PM)	Approach Delay s/veh & LOS AM (PM)	Intersection Delay s/veh & LOS AM (PM)*
Gantzel Rd & Chandler Heights Rd – Signalized	NB	Left	9.23A (14.51B)	22.86C (21.34C)	21.38C (24.16C)
		Thru	23.75C (22.18C)		
		Right	11.52B (15.44B)		
	SB	Left	15.2B (14.8B)	16.35B (26.27C)	
		Thru	16.41B (27.24C)		
		Right	10.77B (12.29B)		
	EB	Left	30.44C (24.49C)	23.27C (22.59C)	
		Thru	23.12C (22.56C)		
		Right	23.12C (22.56C)		
	WB	Left	31.02C (29.65C)	29.22C (26.25C)	
		Thru	27.76C (22.28C)		
		Right	27.76C (22.28C)		
Gantzel Rd & Lone Star Ln – Two-way stop	NB	Left	11.3B (14.48B)	0A (0A)	72.29F (142.46F)
		Thru	0A (0A)		
		Right	0A (0A)		
	SB	Left	14.5B (11.96B)	0.04A (0A)	
		Thru	0A (0A)		
		Right	0A (0A)		
	EB	Left	28.31D (44.48E)	23.36C (44.48E)	
		Thru	35.35E (36.34E)		
		Right	13.46B (17.02C)		
	WB	Left	176.7F (140.2F)	72.29F (142.46F)	
		Thru	236.27F (273.28F)		
		Right	20.09C (13.91B)		
Gantzel Rd & Pecan Creek Dr – Signalized	NB	Left	16.3B (14.16B)	33.75C (18.61B)	30.12C (22.24C)
		Thru	35.75D (19.46B)		
		Right	18.48B (13.99B)		
	SB	Left	22.79C (28.79C)	26.38C (23.27C)	
		Thru	27.26C (22.47C)		
		Right	17.76B (12.86B)		
	EB	Left	17.88B (21.71C)	20.23C (25.01C)	
		Thru	21.71C (28.07C)		
		Right	21.71C (28.07C)		
	WB	Left	17.8B (21.83C)	30.49C (31.27C)	
		Thru	36.72D (35.81D)		
		Right	36.72D (35.81D)		

As shown, the study area intersections of Gantzel Road/Chandler Heights Road and Gantzel Road/Pecan Creek Drive currently operate at acceptable levels of service, LOS C or better, in the morning and evening peak hours. However, the intersection of Gantzel Road/Lone Star Lane operates at an unacceptable level of service, LOS F. This is because motorists wanting to make a left turn from the stop-controlled leg experience delay. Left-turning and through movements on stop-controlled minor roads that intersect with major streets typically experience greater delay for short periods of time in the peak hours due to the wait for acceptable gaps on the major street, while the free-flowing major streets experience minimal delay.

VI. TRAFFIC AND IMPROVEMENT ANALYSIS

The purpose of this section is to show the relations between traffic operations and roadway geometrics; identify needs pertaining to progressive traffic flow and safety; and identify alternatives for further consideration, where applicable.

Within the total traffic volumes figure, the proposed lane configurations are presented. **Figure 7** illustrates the recommendations from this section for the ultimate buildout conditions.

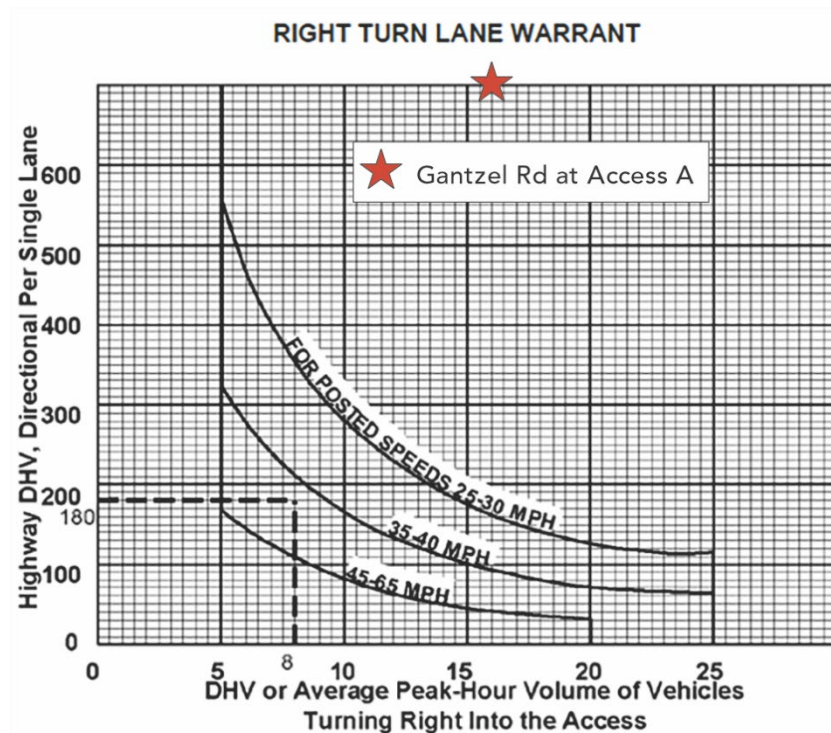
A. Accessibility to Gantzel Commercial Center Development

Access to the Gantzel Commercial Center development is provided via Gantzel Road through a collector road network. The Gantzel Commercial Center Development will have three accesses. Access A is located on Gantzel Road and Accesses B and C are located on Lone Star Lane. Gantzel Road and Lone Star Lane are owned and maintained by Pinal County. Therefore, Accesses A, B and C were analyzed using Pinal County Traffic Engineering Guidelines.

B. Auxiliary Lane Analyses

B.1. Right Turn Lane Warrants

A right-turn deceleration lane is warranted where the intersecting point falls on or above the curve for the posted speed on the Right Turn Lane Warrant graph exhibited below.



Based on the Right Turn Lane Warrant graph, the forecasted right turns into the site Access A (17), the single-lane hourly volume on Gary Road (750 vph), and the posted speed limit (45 mph), a right-turn deceleration lane is warranted on southbound Gary Road at Access A.

B.2. Right Lane Queue and Storage Length Analysis

The lengths of the left-turn and right-turn lanes are designed provide adequate distance for turning vehicles to decelerate and wait to turn without interfering with through traffic. The lengths of left-turn and right-turn lanes adjacent to the proposed development were determined using the total traffic projected for year 2022 and the Pinal County guidelines. The guidelines were applied as follows:

For non-signal controlled intersections:

$$\text{Storage Length} = \frac{\text{vehicles}}{\text{hour}} \times \frac{1 \text{ hour}}{30^*} \times 25 \frac{\text{feet}}{\text{vehicle}}$$

*30 2-minute periods in one hour

All results should be rounded to the nearest 25-foot interval.

The following table summarizes the required lengths for the proposed turn lanes.

Table 8: Right Turn Lane Analysis

Location		Turn Volume 2045	Calculated Queue (feet)	Desirable Braking Distance (feet)	Recommended Storage (feet)	Taper (feet)
On	At					
Pinal County Methodology – Section 5.12 Queuing Analysis						
Gantzel Road	Access A	SB Right: 17	25	-	100	120

C. Traffic Signal Warrant Analysis

The 2009 *Manual on Uniform Traffic Control Devices* (MUTCD) was used as the primary tool to determine if a traffic signal is warranted at the intersection of Gantzel Road/Lone Star Lane and by full buildout.

There are nine specific signal warrants in the MUTCD; however, not all warrants are applicable to this study. The warrants used in this analysis include:

Warrant 1 – Eight-Hour Vehicular Volume

Warrant 2 – Four-Hour Vehicular Volume

Appendix C: Traffic Signal Warrant Analyses presents the results of the signal warrant analyses.

Warrant 1 – Eight-Hour Vehicular Volume

The Minimum Vehicular Volume, Condition A, is intended for application where the volume of intersecting traffic from a side street or driveway is the principal reason for considering installation of a traffic signal. In this condition, the warrant would be satisfied when, for each of any eight hours of an average day, the traffic volumes on the major and minor approach are equal to or exceeds specified limits located on *Table 4C-1 Warrant 1 Eight-Hour Vehicular Volume* in the *MUTCD 2009*.

The Interruption of Continuous Traffic, Condition B, is intended for application where the traffic volume on a major street is so heavy that the traffic on a minor intersection street or driveway has excessive delay or hazard in entering or crossing the major street. This warrant is met when, for each of any eight hours of an average day, the traffic volumes on the major and minor approach is equal to or exceeds specified limits located on *Table 4C-1 Warrant 1, Eight-Hour Vehicular Volume* in the *MUTCD 2009*.

Volume projections for the eight highest hours on an average day were determined by applying hourly adjustment factors calculated from the available peak hour turning movement count data.

Warrant 1 Results:

Gantzel Road/Lone Star Lane - Warrant 1 is not met for both Condition A and Condition B when the developed is constructed.

Warrant 2 – Four-Hour Vehicular Volume

The four-hour vehicular volume signal warrant conditions are intended to be applied where the volume of intersecting traffic is the principal reason to consider installing a traffic control signal. This warrant is satisfied when, for each of any four hours of an average day, plotted points representing vehicles per hour on the major street (total of both approaches) and the vehicles per hour on the minor street approach (one direction only) all fall above the appropriate curve located on *Figure 4C-2 Warrant 2, Four-Hour Vehicular Volume* in the *MUTCD 2009*.

Warrant 2 Results:

Gantzel Road/Lone Star Lane - Warrant 2 is not met when the developed is constructed.

Signal Warrant Summary

The traffic signal warrant analyses within this TIA are based on forecasted traffic volumes and conceptual land plans, site access, and assumed land uses. A traffic signal is not warranted at this time, but traffic signal warrants should be conducted once the development is constructed and operational.

D. Intersection Sight Distance

Proper intersection sight distance and sight triangles shall be provided and maintained at all site access driveways of the proposed development to give drivers exiting the site a clear view of oncoming traffic. The landscape and hardscape within the sight triangles must not obstruct the driver's view of the adjacent travel lanes. To ensure adequate sight distances and sight distance triangles are provided at the site access driveways per the most current edition of the American Association of State Highway and Transportation Officials, A Policy on Geometric Design of Highways and Streets and the Pinal County TIA Guidelines.

To ensure proper sight distance triangles are provided, design plans for landscape and hardscape shall include sight triangle provisions where needed prior to construction. Plans should not be approved that are exempt from these requirements.

Per AASHTO's A Policy on Geometric Design of Highways and Streets manual, 2018, the sight distance along the roadway at the existing driveway locations is determined by the following formula:

$$SD = 1.47 \times V_{\text{major}} \times t_g$$

Where:

- V_{major} = design speed (=posted speed) on Gantzel Road (45 miles per hour)
- t_g = time gap. For crossing a two-lane roadway, the time gap for left-turns (looking right) is 7.5 seconds for passenger vehicles; the time gap for right-turns (looking left) is 6.5 seconds for passenger vehicles.

Assuming a flat grade for the existing driveways, the required sight distance at the site's driveways is as follows and as shown on *Figure 8: Sight Triangle*:

- Gantzel Road - the required sight distance at the site's driveway for right-turns from a stop (looking left) is **335 feet**.
- Gantzel Road – the required sight distance at the site's driveway for left-turns from a stop (looking right) is **386 feet**.

E. Total Traffic

Total traffic projections for the horizon years were determined by adding the proposed development's site's generated traffic to the forecasted background traffic volumes. **Figure 7** presents the total traffic volumes during the morning and evening peak hours and the lane configurations that were modeled for the horizon year.

Figure 7: Total Traffic – Year 2027

Figure 8: Site Triangle

Capacity analyses at the study area intersections and site access intersections were performed using the recommended geometric infrastructure and forecasted total traffic for the horizon year of the study.

Table 9 presents the total levels of service utilizing the recommended improvements. Capacity analyses are in Appendix B.

Table 9: 2027 Total Conditions Intersection Levels of Service

Intersection and Approach		Turning Movement	Delay (s/veh) for Movement & LOS AM (PM)	Approach Delay s/veh & LOS AM (PM)	Intersection Delay s/veh & LOS AM (PM)*
Gantzel Rd & Chandler Heights Rd – Signalized	NB	Left	9.38A (15.08B)	23.32C (21.77C)	21.68C (24.92C)
		Thru	24.26C (22.68C)		
		Right	11.55B (15.5B)		
	SB	Left	15.55B (15.31B)	16.54B (27.48C)	
		Thru	16.59B (28.48C)		
		Right	10.77B (12.29B)		
	EB	Left	30.44C (24.49C)	23.29C (22.64C)	
		Thru	23.15C (22.61C)		
		Right	23.15C (22.61C)		
	WB	Left	31.18C (29.9C)	29.3C (26.43C)	
		Thru	27.76C (22.28C)		
		Right	27.76C (22.28C)		
Gantzel Rd & Lone Star Ln – Two-way stop	NB	Left	11.7B (15.76C)	0.15A (0.38A)	82.63F (182.6F)
		Thru	0A (0A)		
		Right	0A (0A)		
	SB	Left	14.5B (11.96B)	0.04A (0A)	
		Thru	0A (0A)		
		Right	0A (0A)		
	EB	Left	33.18D (71.18F)	28.06D (62.46F)	
		Thru	41.47E (63.96F)		
		Right	17.35C (42.29E)		
	WB	Left	205.46F (183.94F)	82.63F (182.6F)	
		Thru	272.51F (349.95F)		
		Right	21.22C (13.91B)		

Gantzel Rd & Pecan Creek Dr – Signalized	NB	Left	16.46B (14.54B)	34.95C (18.92B)	30.8C (22.76C)
		Thru	37.06D (19.78B)		
		Right	18.48B (13.99B)		
	SB	Left	23.25C (31.05C)	26.72C (24.06C)	
		Thru	27.6C (22.98C)		
		Right	17.78B (12.89B)		
	EB	Left	17.96B (21.79C)	20.23C (24.99C)	
		Thru	21.71C (28.07C)		
		Right	21.71C (28.07C)		
	WB	Left	17.8B (21.83C)	30.75C (31.55C)	
		Thru	37.05D (36.13D)		
		Right	37.05D (36.13D)		
Gantzel Rd & Access A – One-way stop	NB	Thru	0A (0A)	0A (0A)	13.45B (17.37C)
	SB	Thru	0A (0A)	0A (0A)	
		Right	0A (0A)		
	EB	Right	13.45B (17.37C)	13.45B (17.37C)	

As shown, the study area intersections of Gantzel Road/Chandler Heights Road and Gantzel Road/Pecan Creek Drive currently operate at acceptable levels of service, LOS C or better, in the morning and evening peak hours. However, the intersection of Gantzel Road/Lone Star Lane operates at an unacceptable level of service, LOS F. This is because motorists wanting to make a left turn from the stop-controlled leg experience delay. Left-turning and through movements on stop-controlled minor roads that intersect with major streets typically experience greater delay for short periods of time in the peak hours due to the wait for acceptable gaps on the major street, while the free-flowing major streets experience minimal delay.

F. Roadway and Intersection Geometric Improvements

Gantzel Road is noted in the “Regionally Significant Routes for Safety and Mobility, Final Report” dated December 2008 and also shown on the “Regionally Significant Routes for Safety and Mobility-2017 Update” for Pinal County as a principal arterial roadway within the vicinity of the site. Gantzel Road is currently located on the eastern boundary of the Gantzel Commercial Center property.

With the construction of all new accesses, sight triangles shall be provided and maintained at all site accesses per the most current edition of the American Association of State Highway and Transportation Officials, A Policy on Geometric Design of Highways and Streets, and the Pinal County TIA Guidelines. The sight triangles shall be provided on the developer’s landscape and design plans.

Figure 9: Recommendations

VII. CONCLUSIONS AND RECOMMENDATIONS

The approximate 3.3-acre development is planned to include two 10,000 square foot commercial buildings, which will include restaurant and retail spaces. The development will be constructed in one phase with full buildout of Gantzel Commercial Center anticipated by 2027.

The Gantzel Commercial Center Development will have three accesses. Access A, is located on Gantzel Road, Access B and C is located on Lone Star Lane. Gantzel Road and Lone Star Lane are owned and maintained by Pinal County. Therefore, Accesses A, B and C were analyzed using Pinal County Traffic Engineering Guidelines.

On a typical weekday, after full build-out, the Gantzel Commercial Center development is forecasted to generate 1,089 total daily trips, with 79 trips in the morning peak hour and 126 trips in the evening peak hour, according to the 11th Edition of the Trip Generation Manual.

A right-turn lane is warranted on Gantzel Road at Access A, with a 100-foot storage and 120-foot taper.

Traffic signals are not warranted based on traffic generated by the surrounding developments including the Gantzel Commercial Center Development. Signal warrants should be conducted to confirm signals are warranted prior to their installation.

Proper intersection sight distance and sight triangles shall be provided and maintained at all site access driveways of the proposed development to give drivers exiting the site a clear view of oncoming traffic. The landscape and hardscape (monument signs) within the sight triangles must not obstruct the driver's view of the adjacent travel lanes. The sight triangles shall be provided on the developer's landscape and design plans.

VIII. LIMITATIONS

Our professional services have been performed using the degree of skill ordinarily exercised, under similar circumstances, by reputable transportation engineering firms practicing in this locality. No other warranty, expressed or implied, is made.

The contents of this report are intended for the sole use of the addressee and his/her designees. In completing this report, data was obtained from a variety of sources (i.e., City, County, State and Federal sources); United Civil Group has assumed these sources to be reliable and accurate. Should deviations from this report be noted, this firm shall be contacted for review of the area of concern.

A reasonable attempt was made to acquire recent traffic impact studies, traffic projections and/or data that may be helpful in more accurately projecting traffic volumes. United Civil Group is not responsible for incorporating data made available after this document has been finalized.

This report is issued with the understanding that it is the responsibility of the owner to see that its provisions are carried out or brought to the attention of those concerned. If any changes to the proposed project are planned, the conclusions and recommendations contained in this report shall be reviewed and the report shall be modified or supplemented as necessary.

IX. SOURCES

Arizona Department of Transportation Traffic Engineering Guidelines and Processes, June 2015.

A Policy on Geometric Design of Highways and Streets, American Association of State Highway and Transportation Officials, 2018.

Highway Capacity Manual, HCM, Transportation Research Board, 6th Edition.

Manual on Uniform Traffic Control Devices, Federal Highway Administration, MUTCD 2009.

Pinal County Access Management Manual, February 24, 2017.

Pinal County Regionally Significant Routes for Safety and Mobility Final Report. December 2008.

Pinal County Small Area Transportation Study Final Report, August 2006.

Pinal County Subdivision & Infrastructure Design Manual.

Pinal County Traffic Assessment Guidelines & Procedures, January 2007.

Trip Generation, 11th Edition, Institute of Transportation Engineers.

APPENDIX A

Traffic Counts

Project No: TR26013

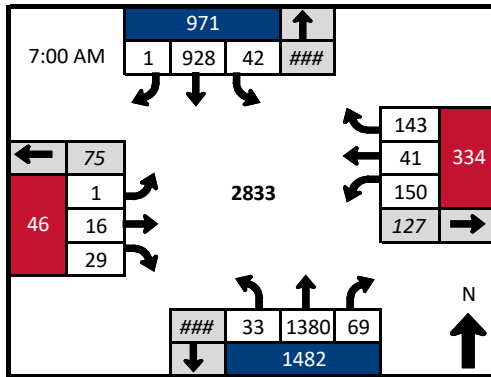
Location: Gantzel Road
and Chandler Heights Road

Intersection Configuration: One-Way Stop Control

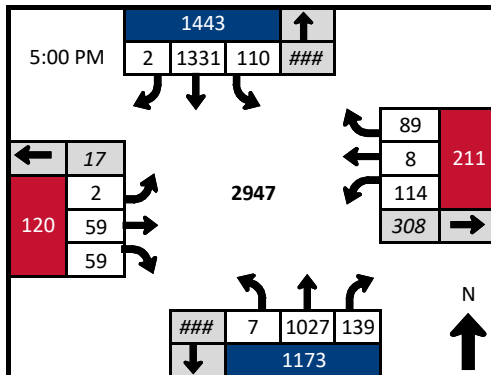
Turning Movement Count

	Speed Limit							
Northbound	45	1		2		1		
Southbound	45	1		2		1		
Eastbound	35	1			1			1
Westbound	35	1			1			

February 11, 2026 (Wednesday)



	Gantzel Road				Gantzel Road				Chandler Heights Road				Chandler Heights Road				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
7:00 AM	10	353	8	0	10	220	0	0	0	1	4	0	36	7	47	0	696	
7:15 AM	4	360	23	0	9	221	1	0	1	5	5	0	35	8	37	0	709	
7:30 AM	8	364	20	0	8	233	0	0	0	2	11	0	44	12	25	0	727	
7:45 AM	11	303	18	0	15	254	0	0	0	8	9	0	35	14	34	0	701	2833
8:00 AM	3	293	21	0	20	248	0	0	1	5	5	0	36	3	29	0	664	2801
8:15 AM	2	268	23	0	7	220	0	0	0	5	8	0	33	4	37	0	607	2699
8:30 AM	4	325	20	0	5	229	0	0	2	1	2	0	30	1	22	0	641	2613
8:45 AM	6	274	18	0	8	158	1	0	1	4	10	0	32	1	18	0	531	2443
Peak Hour Total	33	1380	69	0	42	928	1	0	1	16	29	0	150	41	143	0	2833	
PHF	0.75	0.95	0.75		0.70	0.91	0.25		0.25	0.50	0.66		0.85	0.73	0.76		0.97	



	Gantzel Road				Gantzel Road				Chandler Heights Road				Chandler Heights Road				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
4:00 PM	5	254	26	0	14	330	0	0	0	12	14	0	24	5	24	0	708	
4:15 PM	2	252	30	0	23	314	2	0	1	14	13	0	30	4	19	0	704	
4:30 PM	4	250	37	0	22	332	0	0	0	11	15	0	27	1	21	0	720	
4:45 PM	8	244	26	0	22	299	1	0	2	13	15	0	21	2	21	0	674	2806
5:00 PM	3	271	34	0	29	313	1	0	1	18	17	0	32	5	23	0	747	2845
5:15 PM	1	257	32	0	23	350	0	0	0	15	13	0	26	0	25	0	742	2883
5:30 PM	2	233	33	0	25	333	0	0	0	17	12	0	25	3	25	0	708	2871
5:45 PM	1	266	40	0	33	335	1	0	1	9	17	0	31	0	16	0	750	2947
Peak Hour Total	7	1027	139	0	110	1331	2	0	2	59	59	0	114	8	89	0	2947	
PHF	0.58	0.95	0.87		0.83	0.95	0.50		0.50	0.82	0.87		0.89	0.40	0.89		0.98	

Project No: TR26013

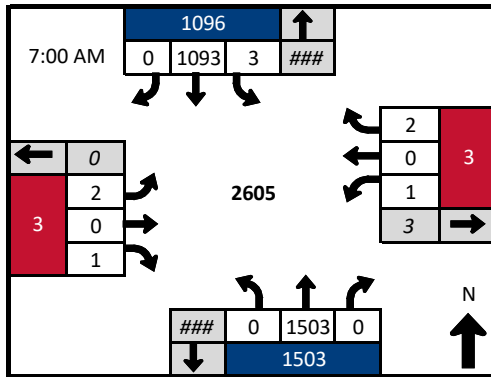
Location: Gantzel Road
and Lone Star Lane

Intersection Configuration: One-Way Stop Control

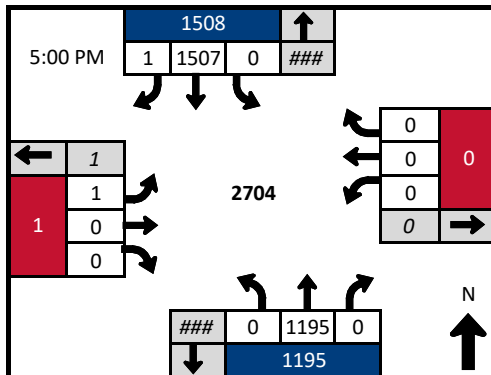
Turning Movement Count

	Speed Limit	Lt	Lt/T	T	T/Rt	Rt	Lt/T/Rt	Lt/Rt
Northbound	45	1		2				
Southbound	45			2		1		
Eastbound	25							1
Westbound								

February 11, 2026 (Wednesday)



	Gantzel Road				Gantzel Road				Lone Star Lane				Lone Star Lane				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
7:00 AM	0	378	0	0	1	253	0	0	1	0	0	0	0	0	0	0	633	
7:15 AM	0	409	0	0	2	264	0	0	0	0	0	0	0	0	0	0	675	
7:30 AM	0	389	0	0	0	290	0	0	0	0	0	0	1	0	1	0	681	
7:45 AM	0	327	0	0	0	286	0	0	1	0	1	0	0	0	1	1	616	2605
8:00 AM	0	326	0	0	1	296	0	0	0	0	0	0	0	0	0	2	623	2595
8:15 AM	1	286	1	0	0	261	1	0	0	0	1	0	0	0	0	0	551	2471
8:30 AM	0	352	0	0	1	262	0	0	0	0	0	0	0	0	0	0	615	2405
8:45 AM	0	297	0	0	0	205	0	0	0	0	0	0	1	0	0	0	503	2292
Peak Hour Total	0	1503	0	0	3	1093	0	0	2	0	1	0	1	0	2	1	2605	
<i>PHF</i>	<i>1.00</i>	<i>0.92</i>	<i>1.00</i>		<i>0.38</i>	<i>0.94</i>	<i>1.00</i>		<i>0.50</i>	<i>1.00</i>	<i>0.25</i>		<i>0.25</i>	<i>1.00</i>	<i>0.50</i>		<i>0.96</i>	



	Gantzel Road				Gantzel Road				Lone Star Lane				Lone Star Lane				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
4:00 PM	0	293	0	0	0	364	1	0	1	0	1	0	0	0	0	3	660	
4:15 PM	0	283	0	0	0	356	1	0	0	0	1	0	0	0	0	0	641	
4:30 PM	0	298	0	0	0	368	0	0	1	0	0	0	0	0	0	0	667	
4:45 PM	1	277	0	0	0	336	0	0	0	0	0	0	0	0	0	0	614	2582
5:00 PM	0	335	0	0	0	367	1	0	0	0	0	0	0	0	0	0	703	2625
5:15 PM	0	302	0	0	0	388	0	0	0	0	0	0	0	0	0	0	690	2674
5:30 PM	0	264	0	0	0	377	0	0	1	0	0	0	0	0	0	0	642	2649
5:45 PM	0	294	0	0	0	375	0	0	0	0	0	0	0	0	0	1	669	2704
Peak Hour Total	0	1195	0	0	0	1507	1	0	1	0	0	0	0	0	0	1	2704	
<i>PHF</i>	<i>1.00</i>	<i>0.89</i>	<i>1.00</i>		<i>1.00</i>	<i>0.97</i>	<i>0.25</i>		<i>0.25</i>	<i>1.00</i>	<i>1.00</i>		<i>1.00</i>	<i>1.00</i>	<i>1.00</i>		<i>0.96</i>	

Project No: TR26013

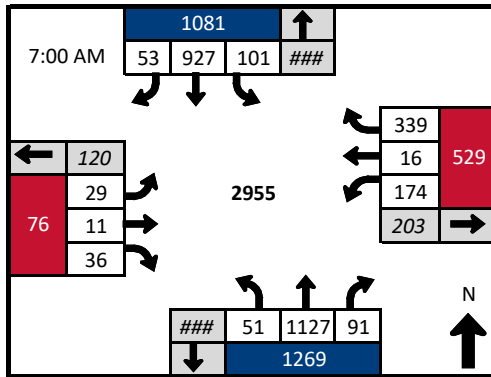
Location: Gantzel Road
and Pecan Creek Drive

Intersection Configuration: One-Way Stop Control

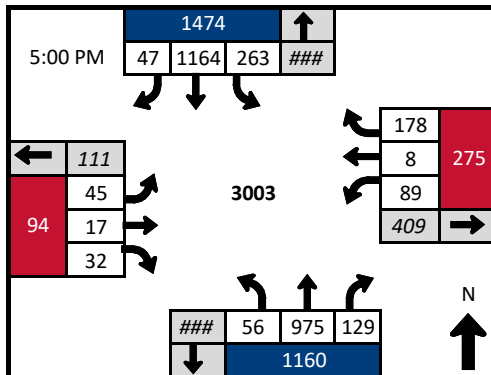
Turning Movement Count

	Speed Limit							
Northbound	45	1		2		1		
Southbound	45	1		2		1		
Eastbound	30	1			1			1
Westbound		1			1			

February 11, 2026 (Wednesday)



	Gantzel Road				Gantzel Road				Pecan Creek Drive				Pecan Creek Drive				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
7:00 AM	11	263	18	0	18	217	13	0	2	1	12	0	36	3	103	0	697	
7:15 AM	12	306	21	1	22	215	10	0	10	3	10	0	36	1	88	0	734	
7:30 AM	12	293	25	0	30	249	19	0	5	2	8	0	48	3	88	0	782	
7:45 AM	16	265	27	0	31	246	11	0	12	5	6	0	54	9	60	0	742	2955
8:00 AM	17	234	19	0	38	236	10	0	12	4	9	0	46	3	59	1	687	2945
8:15 AM	18	238	16	0	35	201	19	0	8	5	9	0	21	3	42	0	615	2826
8:30 AM	13	275	19	0	27	218	14	0	14	2	13	0	20	3	60	0	678	2722
8:45 AM	19	226	25	0	19	171	5	1	14	4	10	0	19	2	59	0	573	2553
Peak Hour Total	51	1127	91	1	101	927	53	0	29	11	36	0	174	16	339	0	2955	
<i>PHF</i>	<i>0.80</i>	<i>0.92</i>	<i>0.84</i>		<i>0.81</i>	<i>0.93</i>	<i>0.70</i>		<i>0.60</i>	<i>0.55</i>	<i>0.75</i>		<i>0.81</i>	<i>0.44</i>	<i>0.82</i>		<i>0.94</i>	



	Gantzel Road				Gantzel Road				Pecan Creek Drive				Pecan Creek Drive				Total	Peak Hour
	Northbound				Southbound				Eastbound				Westbound					
Start Time	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds	Left	Thru	Right	Peds		
4:00 PM	8	229	36	2	65	325	9	0	17	3	9	0	31	3	44	1	779	
4:15 PM	10	232	40	4	72	282	14	0	14	5	7	0	22	1	37	0	736	
4:30 PM	10	245	34	0	66	302	6	1	10	3	13	1	25	1	37	0	752	
4:45 PM	7	224	45	0	50	266	11	0	8	7	7	0	26	3	44	0	698	2965
5:00 PM	10	275	36	0	72	276	15	0	12	5	4	0	20	2	49	0	776	2962
5:15 PM	16	270	20	2	72	300	8	0	7	3	12	0	17	1	36	0	762	2988
5:30 PM	8	206	31	0	65	293	14	0	12	6	6	0	26	3	44	1	714	2950
5:45 PM	22	224	42	0	54	295	10	0	14	3	10	0	26	2	49	0	751	3003
Peak Hour Total	56	975	129	2	263	1164	47	0	45	17	32	0	89	8	178	1	3003	
<i>PHF</i>	<i>0.64</i>	<i>0.89</i>	<i>0.77</i>		<i>0.91</i>	<i>0.97</i>	<i>0.78</i>		<i>0.80</i>	<i>0.71</i>	<i>0.67</i>		<i>0.86</i>	<i>0.67</i>	<i>0.91</i>		<i>0.97</i>	

APPENDIX B

Capacity Analyses

Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	20.9
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.555

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	33	1380	69	42	928	1	1	16	29	150	41	143
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	33	1380	69	42	928	1	1	16	29	150	41	143
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	9	375	19	11	252	0	0	4	8	41	11	39
Total Analysis Volume [veh/h]	36	1500	75	46	1009	1	1	17	32	163	45	155
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	50	0	9	50	0	0	31	0	0	31	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	55	46	46	55	46	46	27	27	27	27
g / C, Green / Cycle	0.61	0.51	0.51	0.61	0.51	0.51	0.30	0.30	0.30	0.30
(v / s)_i Volume / Saturation Flow Rate	0.05	0.42	0.05	0.09	0.28	0.00	0.00	0.03	0.12	0.12
s, saturation flow rate [veh/h]	714	3560	1589	511	3560	1589	1182	1677	1356	1645
c, Capacity [veh/h]	448	1820	812	316	1820	812	294	503	428	494
d1, Uniform Delay [s]	8.74	18.59	11.29	13.62	15.01	10.76	30.22	22.71	28.26	25.10
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.35	4.40	0.23	0.97	1.22	0.00	0.02	0.39	2.56	2.46
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.08	0.82	0.09	0.15	0.55	0.00	0.00	0.10	0.38	0.41
d, Delay for Lane Group [s/veh]	9.10	22.98	11.51	14.59	16.23	10.77	30.24	23.10	30.82	27.56
Lane Group LOS	A	C	B	B	B	B	C	C	C	C
Critical Lane Group	No	Yes	No	Yes	No	No	No	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.29	13.21	0.79	0.41	6.88	0.01	0.02	0.79	3.21	3.67
50th-Percentile Queue Length [ft/ln]	7.37	330.33	19.65	10.20	172.08	0.25	0.48	19.79	80.33	91.63
95th-Percentile Queue Length [veh/ln]	0.53	19.17	1.41	0.73	11.19	0.02	0.03	1.42	5.78	6.60
95th-Percentile Queue Length [ft/ln]	13.27	479.36	35.36	18.36	279.65	0.45	0.87	35.62	144.59	164.94

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	9.10	22.98	11.51	14.59	16.23	10.77	30.24	23.10	23.10	30.82	27.56	27.56
Movement LOS	A	C	B	B	B	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	22.14			16.16			23.24			29.03		
Approach LOS	C			B			C			C		
d_I, Intersection Delay [s/veh]	20.92											
Intersection LOS	C											
Intersection V/C	0.555											

Emissions

Vehicle Miles Traveled [mph]	10.10	420.78	21.04	1.09	23.83	0.02	0.02	1.16	3.87	4.74
Stops [stops/h]	11.80	1057.04	31.43	16.32	550.66	0.40	0.77	31.67	128.53	146.61
Fuel consumption [US gal/h]	0.55	30.18	1.22	0.27	7.36	0.01	0.01	0.45	1.89	2.13
CO [g/h]	38.27	2109.39	84.96	18.98	514.21	0.37	0.80	31.67	132.22	148.69
NOx [g/h]	7.45	410.41	16.53	3.69	100.05	0.07	0.16	6.16	25.73	28.93
VOC [g/h]	8.87	488.87	19.69	4.40	119.17	0.09	0.18	7.34	30.64	34.46

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	1022			1022			600			600		
d_b, Bicycle Delay [s]	10.76			10.76			22.05			22.05		
I_b,int, Bicycle LOS Score for Intersection	2.889			2.431			1.642			2.159		
Bicycle LOS	C			B			A			B		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 2: Gantzel Rd & Lone Star Ln

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 163.6
Level Of Service: F
Volume to Capacity (v/c): 0.043

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1503	0	3	1093	0	2	0	1	1	0	2
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1503	0	3	1093	0	2	0	1	1	0	2
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	408	0	1	297	0	1	0	0	0	0	1
Total Analysis Volume [veh/h]	0	1634	0	3	1188	0	2	0	1	1	0	2
Pedestrian Volume [ped/h]	0			0			0			0		

Intersection Settings				
Priority Scheme	Free	Free	Stop	Stop
Flared Lane			No	No
Storage Area [veh]	0	0	0	0
Two-Stage Gap Acceptance			Yes	No
Number of Storage Spaces in Median	0	0	2	0

Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.00	0.02	0.00	0.01	0.01	0.00	0.01	0.00	0.00	0.04	0.00	0.01
d_M, Delay for Movement [s/veh]	11.17	0.00	0.00	14.23	0.00	0.00	27.53	34.26	13.29	163.56	217.23	19.33
Movement LOS	B	A	A	B	A	A	D	D	B	F	F	C
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.02	0.00	0.00	0.04	0.04	0.04	0.15	0.15	0.15
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.58	0.00	0.00	1.11	1.11	1.11	3.82	3.82	3.82
d_A, Approach Delay [s/veh]	0.00			0.04			22.79			67.41		
Approach LOS	A			A			C			F		
d_I, Intersection Delay [s/veh]	0.11											
Intersection LOS	F											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	29.4
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.633

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	51	1127	91	101	927	53	29	11	36	174	16	339
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	51	1127	91	101	927	53	29	11	36	174	16	339
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	14	306	25	27	252	14	8	3	10	47	4	92
Total Analysis Volume [veh/h]	55	1225	99	110	1008	58	32	12	39	189	17	368
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	10	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	39	0	9	39	0	9	33	0	9	33	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	44	35	35	44	35	35	38	29	38	29
g / C, Green / Cycle	0.49	0.39	0.39	0.49	0.39	0.39	0.42	0.32	0.42	0.32
(v / s)_i Volume / Saturation Flow Rate	0.08	0.34	0.06	0.17	0.28	0.04	0.03	0.03	0.13	0.24
s, saturation flow rate [veh/h]	728	3560	1589	632	3560	1589	1142	1648	1432	1600
c, Capacity [veh/h]	348	1385	618	296	1385	618	388	531	695	516
d1, Uniform Delay [s]	15.06	25.62	17.92	18.75	23.44	17.44	17.44	21.33	16.83	27.22
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.97	8.56	0.55	3.56	3.39	0.30	0.42	0.36	0.96	9.50
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.16	0.88	0.16	0.37	0.73	0.09	0.08	0.10	0.27	0.75
d, Delay for Lane Group [s/veh]	16.02	34.18	18.48	22.31	26.83	17.74	17.86	21.69	17.79	36.72
Lane Group LOS	B	C	B	C	C	B	B	C	B	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.66	13.22	1.40	1.45	9.36	0.80	0.42	0.79	2.63	8.49
50th-Percentile Queue Length [ft/ln]	16.38	330.49	35.09	36.31	234.11	19.94	10.54	19.83	65.76	212.23
95th-Percentile Queue Length [veh/ln]	1.18	19.18	2.53	2.61	14.38	1.44	0.76	1.43	4.73	13.27
95th-Percentile Queue Length [ft/ln]	29.48	479.56	63.17	65.37	359.57	35.90	18.97	35.69	118.37	331.69

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	16.02	34.18	18.48	22.31	26.83	17.74	17.86	21.69	21.69	17.79	36.72	36.72
Movement LOS	B	C	B	C	C	B	B	C	C	B	D	D
d_A, Approach Delay [s/veh]	32.33			25.96			20.21			30.49		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	29.36											
Intersection LOS	C											
Intersection V/C	0.633											

Emissions

Vehicle Miles Traveled [mph]	1.30	28.93	2.34	27.11	248.40	14.29	0.76	1.21	4.48	9.13
Stops [stops/h]	26.20	1057.57	56.15	58.10	749.14	31.91	16.87	31.72	105.22	339.56
Fuel consumption [US gal/h]	0.38	15.56	0.78	1.94	19.87	0.97	0.24	0.45	1.45	5.13
CO [g/h]	26.39	1087.31	54.43	135.35	1388.76	68.09	16.83	31.47	101.38	358.53
NOx [g/h]	5.14	211.55	10.59	26.33	270.20	13.25	3.27	6.12	19.73	69.76
VOC [g/h]	6.12	252.00	12.61	31.37	321.86	15.78	3.90	7.29	23.50	83.09

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	778			778			644			644		
d_b, Bicycle Delay [s]	16.81			16.81			20.67			20.67		
I_b,int, Bicycle LOS Score for Intersection	2.697			2.530			1.697			2.507		
Bicycle LOS	B			B			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	23.6
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.507

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	7	1027	139	110	1331	2	2	59	59	114	8	89
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	7	1027	139	110	1331	2	2	59	59	114	8	89
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	2	279	38	30	362	1	1	16	16	31	2	24
Total Analysis Volume [veh/h]	8	1116	151	120	1447	2	2	64	64	124	9	97
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	45	0	11	47	0	0	34	0	0	34	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	52	41	41	52	43	43	30	30	30	30
g / C, Green / Cycle	0.58	0.46	0.46	0.58	0.48	0.48	0.33	0.33	0.33	0.33
(v / s)_i Volume / Saturation Flow Rate	0.01	0.31	0.09	0.18	0.41	0.00	0.00	0.07	0.10	0.07
s, saturation flow rate [veh/h]	558	3560	1589	670	3560	1589	1288	1718	1262	1610
c, Capacity [veh/h]	310	1622	724	398	1701	759	420	573	405	537
d1, Uniform Delay [s]	13.94	19.43	14.74	12.43	20.67	12.29	24.41	21.61	27.51	21.41
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.15	2.41	0.65	1.94	5.56	0.01	0.02	0.90	1.95	0.82
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.03	0.69	0.21	0.30	0.85	0.00	0.00	0.22	0.31	0.20
d, Delay for Lane Group [s/veh]	14.09	21.84	15.39	14.37	26.24	12.29	24.43	22.51	29.46	22.23
Lane Group LOS	B	C	B	B	C	B	C	C	C	C
Critical Lane Group	Yes	No	No	No	Yes	No	No	No	Yes	No
50th-Percentile Queue Length [veh/ln]	0.07	9.26	1.92	1.20	13.71	0.02	0.03	2.05	2.38	1.69
50th-Percentile Queue Length [ft/ln]	1.87	231.51	48.03	30.04	342.81	0.54	0.84	51.24	59.38	42.17
95th-Percentile Queue Length [veh/ln]	0.13	14.25	3.46	2.16	19.79	0.04	0.06	3.69	4.28	3.04
95th-Percentile Queue Length [ft/ln]	3.37	356.28	86.46	54.07	494.63	0.98	1.51	92.23	106.89	75.90

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	14.09	21.84	15.39	14.37	26.24	12.29	24.43	22.51	22.51	29.46	22.23	22.23
Movement LOS	B	C	B	B	C	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	21.02			25.31			22.54			26.13		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	23.55											
Intersection LOS	C											
Intersection V/C	0.507											

Emissions

Vehicle Miles Traveled [mph]	2.24	313.06	42.36	2.83	34.17	0.05	0.05	3.04	2.94	2.51
Stops [stops/h]	2.99	740.84	76.85	48.07	1096.98	0.87	1.34	81.99	95.02	67.47
Fuel consumption [US gal/h]	0.13	21.94	2.64	0.73	15.19	0.01	0.02	1.16	1.39	0.96
CO [g/h]	9.22	1533.58	184.63	51.25	1062.06	0.82	1.35	81.39	97.13	66.82
NOx [g/h]	1.79	298.38	35.92	9.97	206.64	0.16	0.26	15.84	18.90	13.00
VOC [g/h]	2.14	355.42	42.79	11.88	246.14	0.19	0.31	18.86	22.51	15.49

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	911			956			667			667		
d_b, Bicycle Delay [s]	13.34			12.27			20.00			20.00		
I_b,int, Bicycle LOS Score for Intersection	2.611			2.854			1.774			1.939		
Bicycle LOS	B			C			A			A		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 2: Gantzel Rd & Lone Star Ln

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 42.6
Level Of Service: E
Volume to Capacity (v/c): 0.010

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1195	0	0	1507	1	1	0	0	0	0	0
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1195	0	0	1507	1	1	0	0	0	0	0
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	325	0	0	410	0	0	0	0	0	0	0
Total Analysis Volume [veh/h]	0	1299	0	0	1638	1	1	0	0	0	0	0
Pedestrian Volume [ped/h]	0			0			0			0		

Intersection Settings				
Priority Scheme	Free	Free	Stop	Stop
Flared Lane			No	No
Storage Area [veh]	0	0	0	0
Two-Stage Gap Acceptance			Yes	No
Number of Storage Spaces in Median	0	0	2	0

Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.00	0.01	0.00	0.00	0.02	0.00	0.01	0.00	0.00	0.00	0.00	0.00
d_M, Delay for Movement [s/veh]	14.20	0.00	0.00	11.80	0.00	0.00	42.65	35.19	16.69	130.43	250.24	13.74
Movement LOS	B	A	A	B	A	A	E	E	C	F	F	B
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.03	0.03	0.00	0.00	0.00
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.00	0.00	0.00	0.78	0.78	0.78	0.00	0.00	0.00
d_A, Approach Delay [s/veh]	0.00			0.00			42.65			131.47		
Approach LOS	A			A			E			F		
d_I, Intersection Delay [s/veh]	0.01											
Intersection LOS	E											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	21.9
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.525

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	56	975	129	263	1164	47	45	17	32	89	8	178
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	56	975	129	263	1164	47	45	17	32	89	8	178
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	15	265	35	71	316	13	12	5	9	24	2	48
Total Analysis Volume [veh/h]	61	1060	140	286	1265	51	49	18	35	97	9	193
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	10	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	47	0	9	47	0	9	25	0	9	25	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	52	43	43	52	43	43	30	21	30	21
g / C, Green / Cycle	0.58	0.48	0.48	0.58	0.48	0.48	0.33	0.23	0.33	0.23
(v / s)_i Volume / Saturation Flow Rate	0.10	0.30	0.09	0.44	0.36	0.03	0.04	0.03	0.07	0.13
s, saturation flow rate [veh/h]	601	3560	1589	643	3560	1589	1320	1675	1451	1600
c, Capacity [veh/h]	347	1701	759	388	1701	759	429	391	569	373
d1, Uniform Delay [s]	12.67	17.47	13.46	15.82	19.04	12.68	21.14	27.31	21.18	30.27
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	1.11	1.73	0.53	11.83	2.99	0.17	0.54	0.72	0.65	5.53
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.18	0.62	0.18	0.74	0.74	0.07	0.11	0.14	0.17	0.54
d, Delay for Lane Group [s/veh]	13.78	19.21	13.99	27.65	22.03	12.85	21.68	28.04	21.83	35.81
Lane Group LOS	B	B	B	C	C	B	C	C	C	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.59	8.09	1.67	3.89	10.69	0.57	0.76	0.97	1.52	4.34
50th-Percentile Queue Length [ft/ln]	14.78	202.13	41.86	97.22	267.14	14.30	18.97	24.24	37.91	108.46
95th-Percentile Queue Length [veh/ln]	1.06	12.75	3.01	7.00	16.05	1.03	1.37	1.75	2.73	7.75
95th-Percentile Queue Length [ft/ln]	26.60	318.71	75.35	175.00	401.16	25.74	34.15	43.63	68.24	193.85

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	13.78	19.21	13.99	27.65	22.03	12.85	21.68	28.04	28.04	21.83	35.81	35.81
Movement LOS	B	B	B	C	C	B	C	C	C	C	D	D
d_A, Approach Delay [s/veh]	18.37			22.74			24.98			31.27		
Approach LOS	B			C			C			C		
d_I, Intersection Delay [s/veh]	21.90											
Intersection LOS	C											
Intersection V/C	0.525											

Emissions

Vehicle Miles Traveled [mph]	1.44	25.03	3.31	70.48	311.73	12.57	1.16	1.26	2.30	4.79
Stops [stops/h]	23.64	646.82	66.97	155.55	854.85	22.88	30.35	38.78	60.66	173.53
Fuel consumption [US gal/h]	0.36	8.75	0.90	5.37	23.23	0.78	0.43	0.57	0.86	2.63
CO [g/h]	25.23	611.44	63.25	375.35	1623.52	54.32	30.18	39.73	60.17	183.69
NOx [g/h]	4.91	118.96	12.31	73.03	315.88	10.57	5.87	7.73	11.71	35.74
VOC [g/h]	5.85	141.71	14.66	86.99	376.27	12.59	6.99	9.21	13.94	42.57

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	956			956			467			467		
d_b, Bicycle Delay [s]	12.27			12.27			26.45			26.45		
I_b,int, Bicycle LOS Score for Intersection	2.600			2.881			1.728			2.053		
Bicycle LOS	B			C			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	21.4
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.567

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	33	1380	69	42	928	1	1	16	29	150	41	143
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	34	1408	70	43	947	1	1	16	30	153	42	146
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	9	383	19	12	257	0	0	4	8	42	11	40
Total Analysis Volume [veh/h]	37	1530	76	47	1029	1	1	17	33	166	46	159
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	50	0	9	50	0	0	31	0	0	31	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	55	46	46	55	46	46	27	27	27	27
g / C, Green / Cycle	0.61	0.51	0.51	0.61	0.51	0.51	0.30	0.30	0.30	0.30
(v / s)_i Volume / Saturation Flow Rate	0.05	0.43	0.05	0.09	0.29	0.00	0.00	0.03	0.12	0.12
s, saturation flow rate [veh/h]	704	3560	1589	502	3560	1589	1177	1675	1355	1645
c, Capacity [veh/h]	441	1820	812	310	1820	812	290	502	427	493
d1, Uniform Delay [s]	8.86	18.86	11.30	14.16	15.13	10.76	30.42	22.73	28.37	25.19
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.37	4.89	0.23	1.03	1.28	0.00	0.02	0.40	2.65	2.57
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.08	0.84	0.09	0.15	0.57	0.00	0.00	0.10	0.39	0.42
d, Delay for Lane Group [s/veh]	9.23	23.75	11.52	15.20	16.41	10.77	30.44	23.12	31.02	27.76
Lane Group LOS	A	C	B	B	B	B	C	C	C	C
Critical Lane Group	No	Yes	No	Yes	No	No	No	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.30	13.76	0.80	0.42	7.08	0.01	0.02	0.81	3.29	3.77
50th-Percentile Queue Length [ft/ln]	7.60	344.05	19.92	10.48	177.01	0.25	0.49	20.21	82.17	94.37
95th-Percentile Queue Length [veh/ln]	0.55	19.85	1.43	0.75	11.44	0.02	0.03	1.46	5.92	6.79
95th-Percentile Queue Length [ft/ln]	13.68	496.15	35.86	18.86	286.11	0.45	0.87	36.38	147.91	169.86

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	9.23	23.75	11.52	15.20	16.41	10.77	30.44	23.12	23.12	31.02	27.76	27.76
Movement LOS	A	C	B	B	B	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	22.86			16.35			23.27			29.22		
Approach LOS	C			B			C			C		
d_I, Intersection Delay [s/veh]	21.38											
Intersection LOS	C											
Intersection V/C	0.567											

Emissions

Vehicle Miles Traveled [mph]	10.38	429.20	21.32	1.11	24.30	0.02	0.02	1.19	3.94	4.86
Stops [stops/h]	12.16	1100.97	31.88	16.77	566.44	0.40	0.78	32.34	131.48	150.99
Fuel consumption [US gal/h]	0.56	31.15	1.23	0.28	7.57	0.01	0.01	0.46	1.94	2.19
CO [g/h]	39.42	2177.09	86.12	19.83	528.83	0.37	0.80	32.35	135.36	153.24
NOx [g/h]	7.67	423.58	16.76	3.86	102.89	0.07	0.16	6.29	26.34	29.82
VOC [g/h]	9.14	504.56	19.96	4.60	122.56	0.09	0.19	7.50	31.37	35.52

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	1022			1022			600			600		
d_b, Bicycle Delay [s]	10.76			10.76			22.05			22.05		
I_b,int, Bicycle LOS Score for Intersection	2.915			2.448			1.644			2.172		
Bicycle LOS	C			B			A			B		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 2: Gantzel Rd & Lone Star Ln

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 176.7
Level Of Service: F
Volume to Capacity (v/c): 0.047

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1503	0	3	1093	0	2	0	1	1	0	2
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1533	0	3	1115	0	2	0	1	1	0	2
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	417	0	1	303	0	1	0	0	0	0	1
Total Analysis Volume [veh/h]	0	1666	0	3	1212	0	2	0	1	1	0	2
Pedestrian Volume [ped/h]	0			0			0			0		

Intersection Settings												
Priority Scheme	Free			Free			Stop			Stop		
Flared Lane							No			No		
Storage Area [veh]	0			0			0			0		
Two-Stage Gap Acceptance							Yes			No		
Number of Storage Spaces in Median	0			0			2			0		
Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.00	0.02	0.00	0.01	0.01	0.00	0.01	0.00	0.00	0.05	0.00	0.01
d_M, Delay for Movement [s/veh]	11.30	0.00	0.00	14.50	0.00	0.00	28.31	35.35	13.46	176.70	236.27	20.09
Movement LOS	B	A	A	B	A	A	D	E	B	F	F	C
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.02	0.00	0.00	0.05	0.05	0.05	0.16	0.16	0.16
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.59	0.00	0.00	1.15	1.15	1.15	4.11	4.11	4.11
d_A, Approach Delay [s/veh]	0.00			0.04			23.36			72.29		
Approach LOS	A			A			C			F		
d_I, Intersection Delay [s/veh]	0.11											
Intersection LOS	F											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	30.1
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.640

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	51	1127	91	101	927	53	29	11	36	174	16	339
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	52	1150	91	101	946	54	30	11	37	174	16	339
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	14	313	25	27	257	15	8	3	10	47	4	92
Total Analysis Volume [veh/h]	57	1250	99	110	1028	59	33	12	40	189	17	368
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	10	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	39	0	9	39	0	9	33	0	9	33	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	44	35	35	44	35	35	38	29	38	29
g / C, Green / Cycle	0.49	0.39	0.39	0.49	0.39	0.39	0.42	0.32	0.42	0.32
(v / s)_i Volume / Saturation Flow Rate	0.08	0.35	0.06	0.18	0.29	0.04	0.03	0.03	0.13	0.24
s, saturation flow rate [veh/h]	720	3560	1589	623	3560	1589	1142	1646	1431	1600
c, Capacity [veh/h]	343	1385	618	291	1385	618	388	531	694	516
d1, Uniform Delay [s]	15.26	25.90	17.92	19.06	23.63	17.45	17.45	21.35	16.83	27.22
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	1.05	9.85	0.55	3.72	3.63	0.31	0.43	0.37	0.97	9.50
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.17	0.90	0.16	0.38	0.74	0.10	0.09	0.10	0.27	0.75
d, Delay for Lane Group [s/veh]	16.30	35.75	18.48	22.79	27.26	17.76	17.88	21.71	17.80	36.72
Lane Group LOS	B	D	B	C	C	B	B	C	B	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.68	13.83	1.40	1.46	9.65	0.81	0.44	0.81	2.63	8.49
50th-Percentile Queue Length [ft/ln]	17.05	345.79	35.09	36.51	241.37	20.30	10.88	20.23	65.77	212.23
95th-Percentile Queue Length [veh/ln]	1.23	19.93	2.53	2.63	14.75	1.46	0.78	1.46	4.74	13.27
95th-Percentile Queue Length [ft/ln]	30.69	498.27	63.17	65.72	368.77	36.55	19.58	36.42	118.39	331.69

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	16.30	35.75	18.48	22.79	27.26	17.76	17.88	21.71	21.71	17.80	36.72	36.72
Movement LOS	B	D	B	C	C	B	B	C	C	B	D	D
d_A, Approach Delay [s/veh]	33.75			26.38			20.23			30.49		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	30.12											
Intersection LOS	C											
Intersection V/C	0.640											

Emissions

Vehicle Miles Traveled [mph]	1.35	29.52	2.34	27.11	253.33	14.54	0.78	1.23	4.48	9.13
Stops [stops/h]	27.28	1106.52	56.15	58.42	772.40	32.49	17.41	32.37	105.24	339.56
Fuel consumption [US gal/h]	0.40	16.42	0.78	1.95	20.40	0.99	0.25	0.46	1.45	5.13
CO [g/h]	27.63	1147.92	54.43	136.21	1425.88	69.29	17.37	32.11	101.40	358.53
NOx [g/h]	5.38	223.34	10.59	26.50	277.42	13.48	3.38	6.25	19.73	69.76
VOC [g/h]	6.40	266.04	12.61	31.57	330.46	16.06	4.03	7.44	23.50	83.09

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	778			778			644			644		
d_b, Bicycle Delay [s]	16.81			16.81			20.67			20.67		
I_b,int, Bicycle LOS Score for Intersection	2.720			2.547			1.700			2.507		
Bicycle LOS	B			B			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	24.2
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.517

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	7	1027	139	110	1331	2	2	59	59	114	8	89
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	7	1048	142	112	1358	2	2	60	60	116	8	91
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	2	285	39	30	369	1	1	16	16	32	2	25
Total Analysis Volume [veh/h]	8	1139	154	122	1476	2	2	65	65	126	9	99
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	45	0	11	47	0	0	34	0	0	34	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	52	41	41	52	43	43	30	30	30	30
g / C, Green / Cycle	0.58	0.46	0.46	0.58	0.48	0.48	0.33	0.33	0.33	0.33
(v / s)_i Volume / Saturation Flow Rate	0.01	0.32	0.10	0.18	0.41	0.00	0.00	0.08	0.10	0.07
s, saturation flow rate [veh/h]	549	3560	1589	661	3560	1589	1285	1718	1260	1610
c, Capacity [veh/h]	305	1622	724	392	1701	759	418	573	403	537
d1, Uniform Delay [s]	14.35	19.61	14.77	12.75	20.96	12.29	24.47	21.64	27.64	21.44
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.16	2.57	0.67	2.06	6.28	0.01	0.02	0.92	2.02	0.84
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.03	0.70	0.21	0.31	0.87	0.00	0.00	0.23	0.31	0.20
d, Delay for Lane Group [s/veh]	14.51	22.18	15.44	14.80	27.24	12.29	24.49	22.56	29.65	22.28
Lane Group LOS	B	C	B	B	C	B	C	C	C	C
Critical Lane Group	Yes	No	No	No	Yes	No	No	No	Yes	No
50th-Percentile Queue Length [veh/ln]	0.08	9.56	1.96	1.23	14.30	0.02	0.03	2.08	2.42	1.72
50th-Percentile Queue Length [ft/ln]	1.88	239.03	49.10	30.72	357.62	0.54	0.84	52.12	60.60	43.03
95th-Percentile Queue Length [veh/ln]	0.14	14.63	3.54	2.21	20.51	0.04	0.06	3.75	4.36	3.10
95th-Percentile Queue Length [ft/ln]	3.38	365.80	88.38	55.30	512.69	0.98	1.51	93.81	109.08	77.46

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	14.51	22.18	15.44	14.80	27.24	12.29	24.49	22.56	22.56	29.65	22.28	22.28
Movement LOS	B	C	B	B	C	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	21.34			26.27			22.59			26.25		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	24.16											
Intersection LOS	C											
Intersection V/C	0.517											

Emissions

Vehicle Miles Traveled [mph]	2.24	319.51	43.20	2.88	34.85	0.05	0.05	3.08	2.99	2.56
Stops [stops/h]	3.01	764.88	78.56	49.16	1144.38	0.87	1.34	83.39	96.96	68.85
Fuel consumption [US gal/h]	0.13	22.52	2.70	0.76	15.94	0.01	0.02	1.18	1.42	0.98
CO [g/h]	9.27	1574.18	188.47	52.96	1114.14	0.82	1.35	82.79	99.19	68.19
NOx [g/h]	1.80	306.28	36.67	10.31	216.77	0.16	0.26	16.11	19.30	13.27
VOC [g/h]	2.15	364.83	43.68	12.28	258.21	0.19	0.31	19.19	22.99	15.80

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	911			956			667			667		
d_b, Bicycle Delay [s]	13.34			12.27			20.00			20.00		
I_b,int, Bicycle LOS Score for Intersection	2.633			2.880			1.777			1.946		
Bicycle LOS	B			C			A			A		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 2: Gantzel Rd & Lone Star Ln

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 44.5
Level Of Service: E
Volume to Capacity (v/c): 0.011

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1195	0	0	1507	1	1	0	0	0	0	0
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1219	0	0	1537	1	1	0	0	0	0	0
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	331	0	0	418	0	0	0	0	0	0	0
Total Analysis Volume [veh/h]	0	1325	0	0	1671	1	1	0	0	0	0	0
Pedestrian Volume [ped/h]	0			0			0			0		

Intersection Settings				
Priority Scheme	Free	Free	Stop	Stop
Flared Lane			No	No
Storage Area [veh]	0	0	0	0
Two-Stage Gap Acceptance			Yes	No
Number of Storage Spaces in Median	0	0	2	0

Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.00	0.01	0.00	0.00	0.02	0.00	0.01	0.00	0.00	0.00	0.00	0.00
d_M, Delay for Movement [s/veh]	14.48	0.00	0.00	11.96	0.00	0.00	44.48	36.34	17.02	140.20	273.28	13.91
Movement LOS	B	A	A	B	A	A	E	E	C	F	F	B
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.03	0.03	0.00	0.00	0.00
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.00	0.00	0.00	0.82	0.82	0.82	0.00	0.00	0.00
d_A, Approach Delay [s/veh]	0.00			0.00			44.48			142.46		
Approach LOS	A			A			E			F		
d_I, Intersection Delay [s/veh]	0.01											
Intersection LOS	E											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	22.2
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.531

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	56	975	129	263	1164	47	45	17	32	89	8	178
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	57	995	129	263	1187	48	46	17	33	89	8	178
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	15	270	35	71	323	13	13	5	9	24	2	48
Total Analysis Volume [veh/h]	62	1082	140	286	1290	52	50	18	36	97	9	193
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	10	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	47	0	9	47	0	9	25	0	9	25	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	52	43	43	52	43	43	30	21	30	21
g / C, Green / Cycle	0.58	0.48	0.48	0.58	0.48	0.48	0.33	0.23	0.33	0.23
(v / s)_i Volume / Saturation Flow Rate	0.10	0.30	0.09	0.45	0.36	0.03	0.04	0.03	0.07	0.13
s, saturation flow rate [veh/h]	592	3560	1589	635	3560	1589	1320	1673	1450	1600
c, Capacity [veh/h]	341	1701	759	382	1701	759	429	390	569	373
d1, Uniform Delay [s]	12.99	17.63	13.46	16.20	19.24	12.69	21.15	27.33	21.18	30.27
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	1.17	1.83	0.53	12.59	3.22	0.17	0.55	0.74	0.65	5.53
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.18	0.64	0.18	0.75	0.76	0.07	0.12	0.14	0.17	0.54
d, Delay for Lane Group [s/veh]	14.16	19.46	13.99	28.79	22.47	12.86	21.71	28.07	21.83	35.81
Lane Group LOS	B	B	B	C	C	B	C	C	C	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.60	8.34	1.67	3.95	11.05	0.58	0.77	0.99	1.52	4.34
50th-Percentile Queue Length [ft/ln]	15.09	208.43	41.86	98.77	276.25	14.59	19.37	24.72	37.91	108.46
95th-Percentile Queue Length [veh/ln]	1.09	13.07	3.01	7.11	16.50	1.05	1.39	1.78	2.73	7.75
95th-Percentile Queue Length [ft/ln]	27.16	326.81	75.35	177.79	412.54	26.26	34.87	44.49	68.25	193.85

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	14.16	19.46	13.99	28.79	22.47	12.86	21.71	28.07	28.07	21.83	35.81	35.81
Movement LOS	B	B	B	C	C	B	C	C	C	C	D	D
d_A, Approach Delay [s/veh]	18.61			23.27			25.01			31.27		
Approach LOS	B			C			C			C		
d_I, Intersection Delay [s/veh]	22.24											
Intersection LOS	C											
Intersection V/C	0.531											

Emissions

Vehicle Miles Traveled [mph]	1.46	25.55	3.31	70.48	317.89	12.81	1.19	1.28	2.30	4.79
Stops [stops/h]	24.14	666.97	66.97	158.03	883.99	23.34	31.00	39.55	60.66	173.53
Fuel consumption [US gal/h]	0.37	9.02	0.90	5.45	23.87	0.79	0.44	0.58	0.86	2.63
CO [g/h]	26.02	630.61	63.25	380.94	1668.45	55.40	30.82	40.52	60.18	183.69
NOx [g/h]	5.06	122.69	12.31	74.12	324.62	10.78	6.00	7.88	11.71	35.74
VOC [g/h]	6.03	146.15	14.66	88.29	386.68	12.84	7.14	9.39	13.95	42.57

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft²/ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft²/ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	956			956			467			467		
d_b, Bicycle Delay [s]	12.27			12.27			26.45			26.45		
I_b,int, Bicycle LOS Score for Intersection	2.619			2.903			1.731			2.053		
Bicycle LOS	B			C			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	21.7
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.572

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	33	1380	69	42	928	1	1	16	29	150	41	143
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	1	16	2	0	19	0	0	0	1	2	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	35	1424	72	43	966	1	1	16	31	155	42	146
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	10	387	20	12	263	0	0	4	8	42	11	40
Total Analysis Volume [veh/h]	38	1548	78	47	1050	1	1	17	34	168	46	159
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	50	0	9	50	0	0	31	0	0	31	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	55	46	46	55	46	46	27	27	27	27
g / C, Green / Cycle	0.61	0.51	0.51	0.61	0.51	0.51	0.30	0.30	0.30	0.30
(v / s)_i Volume / Saturation Flow Rate	0.05	0.43	0.05	0.09	0.29	0.00	0.00	0.03	0.12	0.12
s, saturation flow rate [veh/h]	695	3560	1589	497	3560	1589	1177	1673	1353	1645
c, Capacity [veh/h]	435	1820	812	307	1820	812	290	502	426	493
d1, Uniform Delay [s]	8.98	19.03	11.31	14.49	15.25	10.76	30.42	22.74	28.46	25.19
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.40	5.23	0.24	1.06	1.34	0.00	0.02	0.41	2.72	2.57
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.09	0.85	0.10	0.15	0.58	0.00	0.00	0.10	0.39	0.42
d, Delay for Lane Group [s/veh]	9.38	24.26	11.55	15.55	16.59	10.77	30.44	23.15	31.18	27.76
Lane Group LOS	A	C	B	B	B	B	C	C	C	C
Critical Lane Group	No	Yes	No	Yes	No	No	No	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.31	14.11	0.82	0.42	7.29	0.01	0.02	0.83	3.34	3.77
50th-Percentile Queue Length [ft/ln]	7.83	352.71	20.48	10.51	182.29	0.25	0.49	20.63	83.44	94.37
95th-Percentile Queue Length [veh/ln]	0.56	20.27	1.47	0.76	11.72	0.02	0.03	1.49	6.01	6.79
95th-Percentile Queue Length [ft/ln]	14.10	506.71	36.86	18.91	293.00	0.45	0.87	37.14	150.20	169.86

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	9.38	24.26	11.55	15.55	16.59	10.77	30.44	23.15	23.15	31.18	27.76	27.76
Movement LOS	A	C	B	B	B	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	23.32			16.54			23.29			29.30		
Approach LOS	C			B			C			C		
d_I, Intersection Delay [s/veh]	21.68											
Intersection LOS	C											
Intersection V/C	0.572											

Emissions

Vehicle Miles Traveled [mph]	10.66	434.25	21.88	1.11	24.79	0.02	0.02	1.21	3.99	4.86
Stops [stops/h]	12.53	1128.67	32.76	16.81	583.32	0.40	0.78	33.01	133.51	150.99
Fuel consumption [US gal/h]	0.58	31.75	1.27	0.29	7.79	0.01	0.01	0.47	1.97	2.19
CO [g/h]	40.58	2219.56	88.43	20.08	544.48	0.37	0.80	33.02	137.54	153.24
NOx [g/h]	7.90	431.85	17.20	3.91	105.94	0.07	0.16	6.43	26.76	29.82
VOC [g/h]	9.41	514.41	20.49	4.65	126.19	0.09	0.19	7.65	31.88	35.52

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	1022			1022			600			600		
d_b, Bicycle Delay [s]	10.76			10.76			22.05			22.05		
I_b,int, Bicycle LOS Score for Intersection	2.932			2.465			1.645			2.175		
Bicycle LOS	C			B			A			B		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report

Intersection 2: Gantzel Rd & Lone Star Ln

Control Type:	Two-way stop	Delay (sec / veh):	205.5
Analysis Method:	HCM 7th Edition	Level Of Service:	F
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.054

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1503	0	3	1093	0	2	0	1	1	0	2
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	20	0	0	0	11	11	19	0	9	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	20	1533	0	3	1126	11	21	0	10	1	0	2
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	5	417	0	1	306	3	6	0	3	0	0	1
Total Analysis Volume [veh/h]	22	1666	0	3	1224	12	23	0	11	1	0	2
Pedestrian Volume [ped/h]	0			0			0			0		

Priority Scheme	Free	Free	Stop	Stop
Flared Lane			No	No
Storage Area [veh]	0	0	0	0
Two-Stage Gap Acceptance			Yes	No
Number of Storage Spaces in Median	0	0	2	1

Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.04	0.02	0.00	0.01	0.01	0.00	0.15	0.00	0.03	0.05	0.00	0.01
d_M, Delay for Movement [s/veh]	11.70	0.00	0.00	14.50	0.00	0.00	33.18	41.47	17.35	205.46	272.51	21.22
Movement LOS	B	A	A	B	A	A	D	E	C	F	F	C
95th-Percentile Queue Length [veh/ln]	0.12	0.00	0.00	0.02	0.00	0.00	0.63	0.63	0.63	0.19	0.19	0.19
95th-Percentile Queue Length [ft/ln]	3.06	0.00	0.00	0.59	0.00	0.00	15.85	15.85	15.85	4.70	4.70	4.70
d_A, Approach Delay [s/veh]	0.15			0.04			28.06			82.63		
Approach LOS	A			A			D			F		
d_I, Intersection Delay [s/veh]	0.51											
Intersection LOS	F											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	30.8
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.648

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	51	1127	91	101	927	53	29	11	36	174	16	339
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	17	0	2	14	1	1	0	0	0	0	2
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	52	1167	91	103	960	55	31	11	37	174	16	341
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	14	317	25	28	261	15	8	3	10	47	4	93
Total Analysis Volume [veh/h]	57	1268	99	112	1043	60	34	12	40	189	17	371
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	5	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	39	0	9	39	0	9	33	0	9	33	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	44	35	35	44	35	35	38	29	38	29
g / C, Green / Cycle	0.49	0.39	0.39	0.49	0.39	0.39	0.42	0.32	0.42	0.32
(v / s)_i Volume / Saturation Flow Rate	0.08	0.36	0.06	0.18	0.29	0.04	0.03	0.03	0.13	0.24
s, saturation flow rate [veh/h]	713	3560	1589	618	3560	1589	1140	1646	1431	1600
c, Capacity [veh/h]	339	1385	618	287	1385	618	385	531	694	516
d1, Uniform Delay [s]	15.39	26.10	17.92	19.30	23.77	17.46	17.51	21.35	16.83	27.29
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	1.07	10.96	0.55	3.95	3.84	0.31	0.45	0.37	0.97	9.76
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.17	0.92	0.16	0.39	0.75	0.10	0.09	0.10	0.27	0.75
d, Delay for Lane Group [s/veh]	16.46	37.06	18.48	23.25	27.60	17.78	17.96	21.71	17.80	37.05
Lane Group LOS	B	D	B	C	C	B	B	C	B	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.68	14.31	1.40	1.50	9.88	0.83	0.45	0.81	2.63	8.60
50th-Percentile Queue Length [ft/ln]	17.09	357.80	35.09	37.44	246.97	20.66	11.23	20.23	65.77	215.00
95th-Percentile Queue Length [veh/ln]	1.23	20.52	2.53	2.70	15.03	1.49	0.81	1.46	4.74	13.41
95th-Percentile Queue Length [ft/ln]	30.75	512.91	63.17	67.39	375.84	37.19	20.21	36.42	118.39	335.24

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	16.46	37.06	18.48	23.25	27.60	17.78	17.96	21.71	21.71	17.80	37.05	37.05
Movement LOS	B	D	B	C	C	B	B	C	C	B	D	D
d_A, Approach Delay [s/veh]	34.95			26.72			20.23			30.75		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	30.80											
Intersection LOS	C											
Intersection V/C	0.648											

Emissions

Vehicle Miles Traveled [mph]	1.35	29.94	2.34	27.60	257.02	14.79	0.81	1.23	4.48	9.21
Stops [stops/h]	27.34	1144.97	56.15	59.91	790.31	33.06	17.96	32.37	105.24	344.00
Fuel consumption [US gal/h]	0.40	17.12	0.78	2.00	20.81	1.01	0.26	0.46	1.45	5.21
CO [g/h]	27.78	1196.85	54.43	139.59	1454.35	70.49	17.95	32.11	101.40	363.84
NOx [g/h]	5.40	232.86	10.59	27.16	282.96	13.71	3.49	6.25	19.73	70.79
VOC [g/h]	6.44	277.38	12.61	32.35	337.06	16.34	4.16	7.44	23.50	84.32

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	778			778			644			644		
d_b, Bicycle Delay [s]	16.81			16.81			20.67			20.67		
I_b,int, Bicycle LOS Score for Intersection	2.734			2.562			1.702			2.512		
Bicycle LOS	B			B			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 4: Gantzel Rd & Access A

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 13.4
Level Of Service: B
Volume to Capacity (v/c): 0.023

Intersection Setup

Name	Gantzel Rd		Gantzel Rd		Access A	
Approach	Northbound		Southbound		Eastbound	
Lane Configuration						
Turning Movement	Left	Thru	Thru	Right	Left	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	0	0	0	1	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00		30.00		30.00	
Grade [%]	0.00		0.00		0.00	
Crosswalk	No		No		No	

Volumes

Name	Gantzel Rd		Gantzel Rd		Access A	
Base Volume Input [veh/h]	0	1503	1095	0	0	0
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0000	1.0200	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	20	9	11	0	9
Diverted Trips [veh/h]	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1553	1126	11	0	9
Peak Hour Factor	1.0000	0.9200	0.9200	0.9200	1.0000	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	422	306	3	0	2
Total Analysis Volume [veh/h]	0	1688	1224	12	0	10
Pedestrian Volume [ped/h]	0		0		0	

Intersection Settings			
Priority Scheme	Free	Free	Stop
Flared Lane			
Storage Area [veh]	0	0	0
Two-Stage Gap Acceptance			No
Number of Storage Spaces in Median	0	0	0

Movement, Approach, & Intersection Results						
V/C, Movement V/C Ratio	0.00	0.02	0.01	0.00	0.00	0.02
d_M, Delay for Movement [s/veh]	0.00	0.00	0.00	0.00	0.00	13.45
Movement LOS		A	A	A		B
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.00	0.00	0.07
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.00	0.00	1.76
d_A, Approach Delay [s/veh]	0.00		0.00		13.45	
Approach LOS	A		A		B	
d_I, Intersection Delay [s/veh]	0.05					
Intersection LOS	B					

Intersection Level Of Service Report
Intersection 1: Gantzel Rd & Chandler Heights Rd

Control Type:	Signalized	Delay (sec / veh):	24.9
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.529

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	110.00	100.00	110.00	110.00	100.00	110.00	150.00	100.00	100.00	200.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Chandler Heights Rd			Chandler Heights Rd		
Base Volume Input [veh/h]	7	1027	139	110	1331	2	2	59	59	114	8	89
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200	1.0200
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	2	28	3	0	28	0	0	0	2	3	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	9	1076	145	112	1386	2	2	60	62	119	8	91
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	2	292	39	30	377	1	1	16	17	32	2	25
Total Analysis Volume [veh/h]	10	1170	158	122	1507	2	2	65	67	129	9	99
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss	Permiss
Signal Group	1	6	0	5	2	0	0	8	0	0	4	0
Auxiliary Signal Groups												
Maximum Green [s]	16	17	0	15	16	0	0	16	0	0	16	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	0.0	2.0	0.0	0.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	45	0	11	47	0	0	34	0	0	34	0
Lead / Lag	Lead	-	-	Lead	-	-	-	-	-	-	-	-
Minimum Green [s]	5	10	0	5	10	0	0	10	0	0	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	0.0	3.0	0.0	0.0	3.0	0.0
Minimum Recall	No	No		No	No			No			No	
Maximum Recall	No	No		No	No			No			No	
Pedestrian Recall	No	No		No	No			No			No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	2.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	2.00	2.00	2.00	2.00
g_i, Effective Green Time [s]	52	41	41	52	43	43	30	30	30	30
g / C, Green / Cycle	0.58	0.46	0.46	0.58	0.48	0.48	0.33	0.33	0.33	0.33
(v / s)_i Volume / Saturation Flow Rate	0.02	0.33	0.10	0.19	0.42	0.00	0.00	0.08	0.10	0.07
s, saturation flow rate [veh/h]	540	3560	1589	649	3560	1589	1285	1716	1258	1610
c, Capacity [veh/h]	299	1622	724	385	1701	759	418	572	401	537
d1, Uniform Delay [s]	14.87	19.87	14.81	13.15	21.28	12.29	24.47	21.67	27.79	21.44
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	0.21	2.81	0.69	2.16	7.20	0.01	0.02	0.94	2.11	0.84
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.03	0.72	0.22	0.32	0.89	0.00	0.00	0.23	0.32	0.20
d, Delay for Lane Group [s/veh]	15.08	22.68	15.50	15.31	28.48	12.29	24.49	22.61	29.90	22.28
Lane Group LOS	B	C	B	B	C	B	C	C	C	C
Critical Lane Group	Yes	No	No	No	Yes	No	No	No	Yes	No
50th-Percentile Queue Length [veh/ln]	0.09	9.98	2.02	1.24	14.99	0.02	0.03	2.12	2.50	1.72
50th-Percentile Queue Length [ft/ln]	2.36	249.49	50.53	30.88	374.77	0.54	0.84	53.01	62.39	43.03
95th-Percentile Queue Length [veh/ln]	0.17	15.16	3.64	2.22	21.34	0.04	0.06	3.82	4.49	3.10
95th-Percentile Queue Length [ft/ln]	4.25	379.01	90.95	55.58	533.52	0.98	1.51	95.42	112.31	77.46

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	15.08	22.68	15.50	15.31	28.48	12.29	24.49	22.61	22.61	29.90	22.28	22.28
Movement LOS	B	C	B	B	C	B	C	C	C	C	C	C
d_A, Approach Delay [s/veh]	21.77			27.48			22.64			26.43		
Approach LOS	C			C			C			C		
d_I, Intersection Delay [s/veh]	24.92											
Intersection LOS	C											
Intersection V/C	0.529											

Emissions

Vehicle Miles Traveled [mph]	2.81	328.21	44.32	2.88	35.59	0.05	0.05	3.13	3.06	2.56
Stops [stops/h]	3.78	798.37	80.85	49.41	1199.27	0.87	1.34	84.82	99.83	68.85
Fuel consumption [US gal/h]	0.17	23.32	2.77	0.77	16.83	0.01	0.02	1.20	1.46	0.98
CO [g/h]	11.68	1630.18	193.61	53.93	1176.13	0.82	1.35	84.22	102.22	68.19
NOx [g/h]	2.27	317.17	37.67	10.49	228.83	0.16	0.26	16.39	19.89	13.27
VOC [g/h]	2.71	377.81	44.87	12.50	272.58	0.19	0.31	19.52	23.69	15.80

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	911			956			667			667		
d_b, Bicycle Delay [s]	13.34			12.27			20.00			20.00		
I_b,int, Bicycle LOS Score for Intersection	2.663			2.905			1.781			1.951		
Bicycle LOS	B			C			A			A		

Sequence

Ring 1	1	2	-	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	-	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report

Intersection 2: Gantzel Rd & Lone Star Ln

Control Type:	Two-way stop	Delay (sec / veh):	71.2
Analysis Method:	HCM 7th Edition	Level Of Service:	F
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.418

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	0	1	0	1	0	0	0	0	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Lone Star In			Lone Star In		
Base Volume Input [veh/h]	0	1195	0	0	1507	1	1	0	0	0	0	0
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	30	0	0	0	17	17	33	0	15	0	0	0
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	30	1219	0	0	1554	18	34	0	15	0	0	0
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	8	331	0	0	422	5	9	0	4	0	0	0
Total Analysis Volume [veh/h]	33	1325	0	0	1689	20	37	0	16	0	0	0
Pedestrian Volume [ped/h]	0			0			0			0		

Priority Scheme	Free	Free	Stop	Stop
Flared Lane			No	No
Storage Area [veh]	0	0	0	0
Two-Stage Gap Acceptance			Yes	No
Number of Storage Spaces in Median	0	0	2	0

Movement, Approach, & Intersection Results												
V/C, Movement V/C Ratio	0.09	0.01	0.00	0.00	0.02	0.00	0.42	0.00	0.05	0.00	0.00	0.00
d_M, Delay for Movement [s/veh]	15.76	0.00	0.00	11.96	0.00	0.00	71.18	63.96	42.29	183.94	349.95	13.91
Movement LOS	C	A	A	B	A	A	F	F	E	F	F	B
95th-Percentile Queue Length [veh/ln]	0.29	0.00	0.00	0.00	0.00	0.00	2.08	2.08	2.08	0.00	0.00	0.00
95th-Percentile Queue Length [ft/ln]	7.35	0.00	0.00	0.00	0.00	0.00	52.00	52.00	52.00	0.00	0.00	0.00
d_A, Approach Delay [s/veh]	0.38			0.00			62.46			182.60		
Approach LOS	A			A			F			F		
d_I, Intersection Delay [s/veh]	1.23											
Intersection LOS	F											

Intersection Level Of Service Report
Intersection 3: Gantzel Rd & Pecan Creek Dr

Control Type:	Signalized	Delay (sec / veh):	22.8
Analysis Method:	HCM 7th Edition	Level Of Service:	C
Analysis Period:	15 minutes	Volume to Capacity (v/c):	0.543

Intersection Setup

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Approach	Northbound			Southbound			Eastbound			Westbound		
Lane Configuration												
Turning Movement	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right	Left	Thru	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	1	0	1	1	0	1	1	0	0	1	0	0
Entry Pocket Length [ft]	200.00	100.00	125.00	190.00	100.00	70.00	115.00	100.00	100.00	110.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00			30.00			30.00			30.00		
Grade [%]	0.00			0.00			0.00			0.00		
Curb Present	No			No			No			No		
Crosswalk	No			No			No			No		

Volumes

Name	Gantzel Rd			Gantzel Rd			Pecan Creek Dr			Pecan Creek Dr		
Base Volume Input [veh/h]	56	975	129	263	1164	47	45	17	32	89	8	178
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Proportion of CAVs [%]	0.00											
Growth Factor	1.0200	1.0200	1.0000	1.0000	1.0200	1.0200	1.0200	1.0000	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	25	0	3	25	2	2	0	0	0	0	3
Diverted Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Right Turn on Red Volume [veh/h]	0	0	0	0	0	0	0	0	0	0	0	0
Total Hourly Volume [veh/h]	57	1020	129	266	1212	50	48	17	33	89	8	181
Peak Hour Factor	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	15	277	35	72	329	14	13	5	9	24	2	49
Total Analysis Volume [veh/h]	62	1109	140	289	1317	54	52	18	36	97	9	197
Presence of On-Street Parking	No		No	No		No	No		No	No		No
On-Street Parking Maneuver Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
Local Bus Stopping Rate [/h]	0	0	0	0	0	0	0	0	0	0	0	0
v_do, Outbound Pedestrian Volume crossing	0			0			0			0		
v_di, Inbound Pedestrian Volume crossing m	0			0			0			0		
v_co, Outbound Pedestrian Volume crossing	0			0			0			0		
v_ci, Inbound Pedestrian Volume crossing mi	0			0			0			0		
v_ab, Corner Pedestrian Volume [ped/h]	0			0			0			0		
Bicycle Volume [bicycles/h]	0			0			0			0		

Intersection Settings

Located in CBD	No
Signal Coordination Group	-
Cycle Length [s]	90
Active Pattern	Pattern 1
Coordination Type	Time of Day Pattern Isolated
Actuation Type	Fixed time
Offset [s]	0.0
Offset Reference	Lead Green - Beginning of First Green
Permissive Mode	SingleBand
Lost time [s]	0.00

Phasing & Timing (Basic)

Control Type	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss	ProtPer	Permiss	Permiss
Signal Group	1	6	0	5	2	0	3	8	0	7	4	0
Auxiliary Signal Groups												
Maximum Green [s]	11	12	0	10	11	0	11	12	0	10	11	0
Amber [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
All red [s]	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0	1.0	1.0	0.0
Walk [s]	0	5	0	0	5	0	0	5	0	0	5	0
Pedestrian Clearance [s]	0	10	0	0	10	0	0	10	0	0	10	0
Delayed Vehicle Green [s]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Rest In Walk		No			No			No			No	
I1, Start-Up Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
I2, Clearance Lost Time [s]	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0	2.0	2.0	0.0
Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Location [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Advanced Detector Length [ft]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
I, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Phasing & Timing: Pattern 1

Split [s]	9	47	0	9	47	0	9	25	0	9	25	0
Lead / Lag	Lead	-	-	Lead	-	-	Lead	-	-	Lead	-	-
Minimum Green [s]	5	10	0	5	10	0	5	10	0	5	10	0
Vehicle Extension [s]	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0	3.0	3.0	0.0
Minimum Recall	No	No		No	No		No	No		No	No	
Maximum Recall	No	No		No	No		No	No		No	No	
Pedestrian Recall	No	No		No	No		No	No		No	No	

Exclusive Pedestrian Phase

Pedestrian Signal Group	0
Pedestrian Walk [s]	0
Pedestrian Clearance [s]	0

Lane Group Calculations

Lane Group	L	C	R	L	C	R	L	C	L	C
C, Calculated Cycle Length [s]	90	90	90	90	90	90	90	90	90	90
L, Total Lost Time per Cycle [s]	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00
l1_p, Permitted Start-Up Lost Time [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
l2, Clearance Lost Time [s]	0.00	2.00	2.00	0.00	2.00	2.00	0.00	2.00	0.00	2.00
g_i, Effective Green Time [s]	52	43	43	52	43	43	30	21	30	21
g / C, Green / Cycle	0.58	0.48	0.48	0.58	0.48	0.48	0.33	0.23	0.33	0.23
(v / s)_i Volume / Saturation Flow Rate	0.11	0.31	0.09	0.46	0.37	0.03	0.04	0.03	0.07	0.13
s, saturation flow rate [veh/h]	582	3560	1589	625	3560	1589	1317	1673	1450	1600
c, Capacity [veh/h]	335	1701	759	375	1701	759	425	390	569	373
d1, Uniform Delay [s]	13.33	17.82	13.46	16.94	19.48	12.70	21.21	27.33	21.18	30.36
k, delay calibration	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
l, Upstream Filtering Factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
d2, Incremental Delay [s]	1.22	1.96	0.53	14.11	3.51	0.18	0.59	0.74	0.65	5.77
d3, Initial Queue Delay [s]	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Rp, platoon ratio	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PF, progression factor	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Lane Group Results

X, volume / capacity	0.19	0.65	0.18	0.77	0.77	0.07	0.12	0.14	0.17	0.55
d, Delay for Lane Group [s/veh]	14.54	19.78	13.99	31.05	22.98	12.89	21.79	28.07	21.83	36.13
Lane Group LOS	B	B	B	C	C	B	C	C	C	D
Critical Lane Group	No	Yes	No	Yes	No	No	Yes	No	No	Yes
50th-Percentile Queue Length [veh/ln]	0.61	8.65	1.67	4.12	11.46	0.61	0.81	0.99	1.52	4.45
50th-Percentile Queue Length [ft/ln]	15.15	216.35	41.86	102.95	286.46	15.17	20.19	24.72	37.91	111.21
95th-Percentile Queue Length [veh/ln]	1.09	13.48	3.01	7.41	17.01	1.09	1.45	1.78	2.73	7.91
95th-Percentile Queue Length [ft/ln]	27.27	336.96	75.35	185.31	425.25	27.31	36.35	44.49	68.25	197.68

Movement, Approach, & Intersection Results

d_M, Delay for Movement [s/veh]	14.54	19.78	13.99	31.05	22.98	12.89	21.79	28.07	28.07	21.83	36.13	36.13
Movement LOS	B	B	B	C	C	B	C	C	C	C	D	D
d_A, Approach Delay [s/veh]	18.92			24.06			24.99			31.55		
Approach LOS	B			C			C			C		
d_I, Intersection Delay [s/veh]	22.76											
Intersection LOS	C											
Intersection V/C	0.543											

Emissions

Vehicle Miles Traveled [mph]	1.46	26.19	3.31	71.22	324.54	13.31	1.23	1.28	2.30	4.89
Stops [stops/h]	24.24	692.32	66.97	164.72	916.67	24.27	32.31	39.55	60.66	177.93
Fuel consumption [US gal/h]	0.38	9.37	0.90	5.67	24.58	0.82	0.46	0.58	0.86	2.70
CO [g/h]	26.40	654.79	63.25	396.16	1718.48	57.56	32.15	40.52	60.18	188.65
NOx [g/h]	5.14	127.40	12.31	77.08	334.35	11.20	6.26	7.88	11.71	36.70
VOC [g/h]	6.12	151.75	14.66	91.81	398.27	13.34	7.45	9.39	13.95	43.72

Other Modes

g_Walk,mi, Effective Walk Time [s]	0.0			0.0			0.0			0.0		
M_corner, Corner Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
M_CW, Crosswalk Circulation Area [ft ² /ped]	0.00			0.00			0.00			0.00		
d_p, Pedestrian Delay [s]	0.00			0.00			0.00			0.00		
I_p,int, Pedestrian LOS Score for Intersectio	0.000			0.000			0.000			0.000		
Crosswalk LOS	F			F			F			F		
s_b, Saturation Flow Rate of the bicycle lane	2000			2000			2000			2000		
c_b, Capacity of the bicycle lane [bicycles/h]	956			956			467			467		
d_b, Bicycle Delay [s]	12.27			12.27			26.45			26.45		
I_b,int, Bicycle LOS Score for Intersection	2.641			2.929			1.735			2.060		
Bicycle LOS	B			C			A			B		

Sequence

Ring 1	1	2	3	4	-	-	-	-	-	-	-	-	-	-	-	-
Ring 2	5	6	7	8	-	-	-	-	-	-	-	-	-	-	-	-
Ring 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ring 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



Intersection Level Of Service Report
Intersection 4: Gantzel Rd & Access A

Control Type: Two-way stop
Analysis Method: HCM 7th Edition
Analysis Period: 15 minutes

Delay (sec / veh): 17.4
Level Of Service: C
Volume to Capacity (v/c): 0.052

Intersection Setup

Name	Gantzel Rd		Gantzel Rd		Access A	
Approach	Northbound		Southbound		Eastbound	
Lane Configuration						
Turning Movement	Left	Thru	Thru	Right	Left	Right
Lane Width [ft]	12.00	12.00	12.00	12.00	12.00	12.00
No. of Lanes in Entry Pocket	0	0	0	1	0	0
Entry Pocket Length [ft]	100.00	100.00	100.00	100.00	100.00	100.00
No. of Lanes in Exit Pocket	0	0	0	0	0	0
Exit Pocket Length [ft]	0.00	0.00	0.00	0.00	0.00	0.00
Speed [mph]	30.00		30.00		30.00	
Grade [%]	0.00		0.00		0.00	
Crosswalk	No		No		No	

Volumes

Name	Gantzel Rd		Gantzel Rd		Access A	
Base Volume Input [veh/h]	0	1195	1507	0	0	0
Base Volume Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Heavy Vehicles Percentage [%]	2.00	2.00	2.00	2.00	2.00	2.00
Growth Factor	1.0000	1.0200	1.0200	1.0000	1.0000	1.0000
In-Process Volume [veh/h]	0	0	0	0	0	0
Site-Generated Trips [veh/h]	0	30	15	17	0	15
Diverted Trips [veh/h]	0	0	0	0	0	0
Pass-by Trips [veh/h]	0	0	0	0	0	0
Existing Site Adjustment Volume [veh/h]	0	0	0	0	0	0
Other Volume [veh/h]	0	0	0	0	0	0
Total Hourly Volume [veh/h]	0	1249	1552	17	0	15
Peak Hour Factor	1.0000	0.9200	0.9200	0.9200	1.0000	0.9200
Other Adjustment Factor	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Total 15-Minute Volume [veh/h]	0	339	422	5	0	4
Total Analysis Volume [veh/h]	0	1358	1687	18	0	16
Pedestrian Volume [ped/h]	0		0		0	

Intersection Settings			
Priority Scheme	Free	Free	Stop
Flared Lane			
Storage Area [veh]	0	0	0
Two-Stage Gap Acceptance			No
Number of Storage Spaces in Median	0	0	0

Movement, Approach, & Intersection Results						
V/C, Movement V/C Ratio	0.00	0.01	0.02	0.00	0.00	0.05
d_M, Delay for Movement [s/veh]	0.00	0.00	0.00	0.00	0.00	17.37
Movement LOS		A	A	A		C
95th-Percentile Queue Length [veh/ln]	0.00	0.00	0.00	0.00	0.00	0.16
95th-Percentile Queue Length [ft/ln]	0.00	0.00	0.00	0.00	0.00	4.11
d_A, Approach Delay [s/veh]	0.00		0.00		17.37	
Approach LOS	A		A		C	
d_I, Intersection Delay [s/veh]	0.09					
Intersection LOS	C					

WATER AND WASTEWATER DEMAND ANALYSIS

Ganztel Commercial Center

Pinal County, Arizona (San Tan Valley Service Area)

1.0 PROJECT DESCRIPTION

The proposed project consists of a **19,000 square foot commercial center** located within the San Tan Valley service area. The development program includes:

- **3,000 SF** – Restaurant space
- **16,000 SF** – Retail and professional office uses

This analysis provides an estimate of **average daily water demand, wastewater generation, and peak demand conditions** for planning and preliminary design purposes.

2.0 DESIGN CRITERIA

Water demand and wastewater flow projections are based on planning-level unit demand factors consistent with standards utilized by EPCOR Water Arizona Inc. for commercial developments in Pinal County.

2.1 Unit Demand Factors

<u>Land Use</u>	<u>Water Demand</u>	<u>Sewer Return Factor</u>
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<u>Restaurant</u>	<u>0.30 gpd/sf</u>	<u>95%</u>
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<u>Retail / Office</u>	<u>0.08 gpd/sf</u>	<u>85%</u>
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2.2 Peaking Factors

<u>Parameter</u>	<u>Factor</u>
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<u>Maximum Day Demand (MDD)</u>	<u>2.25 × ADF</u>
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<u>Peak Hour Demand (PHD)</u>	<u>3.5 × ADF</u>
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3.0 WATER DEMAND ANALYSIS

3.1 Average Daily Demand (ADF)

Restaurant Use:

$$3,000 \text{ SF} \times 0.30 \frac{\text{gpd}}{\text{SF}} = 900 \text{ gpd}$$

Retail / Office Use:

$$16,000 \text{ SF} \times 0.08 \frac{\text{gpd}}{\text{SF}} = 1,280 \text{ gpd}$$

Total Average Daily Demand:

$$ADF = 900 + 1,280 = 2,180 \text{ gpd}$$

3.2 Peak Demand Conditions

Maximum Day Demand (MDD):

$$MDD = 2,180 \times 2.25 = 4,905 \text{ gpd}$$

Peak Hour Demand (PHD):

$$PHD = 2,180 \times 3.5 = 7,630 \text{ gpd}$$

4.0 WASTEWATER FLOW ANALYSIS

Wastewater flows are estimated by applying return factors to the projected water demand.

4.1 Average Daily Sewer Flow

Restaurant Use:

$$900 \times 0.95 = 855 \text{ gpd}$$

Retail / Office Use:

$$1,280 \times 0.85 = 1,088 \text{ gpd}$$

Total Average Daily Sewer Flow:

$$ADF_{\text{sewer}} = 855 + 1,088 = 1,943 \text{ gpd}$$

5.0 SUMMARY OF DEMANDS

Parameter

Flow

Average Daily Water Demand (ADF) 2,180 gpd

<u>Parameter</u>	<u>Flow</u>
<u>Maximum Day Demand (MDD)</u>	<u>4,905 gpd</u>
<u>Peak Hour Demand (PHD)</u>	<u>7,630 gpd</u>
<u>Average Daily Sewer Flow</u>	<u>1,943 gpd</u>

6.0 DESIGN CONSIDERATIONS

The following items shall be addressed during final design and agency review:

- **Water Meter Sizing:** To be determined based on peak hour demand and fixture unit analysis (anticipated range: 1.5"–2" meter).
 - **Fire Flow Requirements:** To be established per applicable fire code; not included in this analysis.
 - **Irrigation Demand:** Landscape water use to be metered separately and is excluded from this calculation.
 - **Grease Interceptor:** Required for restaurant use per local code.
 - **Fixture Unit Verification:** Final demand subject to confirmation based on tenant improvements and plumbing fixture counts.
 - **Sewer Infrastructure:** Connection to public sewer system subject to minimum pipe sizing and utility standards.
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7.0 LIMITATIONS

This analysis is intended for **planning and preliminary engineering purposes only**. Final water and wastewater demands are subject to revision based on:

- Tenant-specific uses
- Detailed plumbing plans
- Utility provider review and approval
- Applicable local and state regulations